

PROBABILITY III

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Chapter 1

REFINING THE NOTION OF PROBABILITY

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Consider the random walk on the line; a drunk ant sits at the origin and takes steps of length 1 to the left or right with equal probability. We wish to obtain the probability that after an even n steps, the ant returns to the origin. Of course, the natural approach is to frame this problem in a discrete probability space.

Definition 1.1. A *discrete probability space* is a pair (Ω, q) where Ω is a non-empty finite or countably infinite set called the sample space, and $q : \Omega \rightarrow [0, 1]$ is a function satisfying $\sum_{\omega \in \Omega} q(\omega) = 1$. Any subset of Ω is termed an event, and the probability of an event $A \subseteq \Omega$ is defined as $\mathbb{P}(A) = \sum_{\omega \in A} q(\omega)$.

For the problem above, we can take the sample space to be

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots, x_n) \in \mathbb{Z}^{n+1} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } 0 \leq i < n\}, \quad (1.1)$$

and the mapping q to be $q(\mathbf{x}) = 2^{-n}$ for all $\mathbf{x} \in \Omega$. Then, the event that the ant returns to the origin after n steps is given by $A = \{\mathbf{x} \in \Omega \mid x_n = 0\}$, and we have

$$\mathbb{P}(A) = \sum_{\mathbf{x} \in A} q(\mathbf{x}) = 2^{-n} |A| = 2^{-n} \binom{n}{n/2}. \quad (1.2)$$

Reframing the problem, let us consider a random walk on the line starting at the origin, and continuing indefinitely. In this case, the sample space is given by

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots) \in \mathbb{Z}^{\mathbb{N}} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } i \in \mathbb{N}\}. \quad (1.3)$$

This Ω is uncountably infinite. In such a case, we cannot define a probability measure by assigning a positive probability to each element of Ω . As another example of difficulty in dealing with uncountable sample spaces, consider the problem of choosing a real number uniformly at random from the interval $[0, 1]$. If we attempt to assign a positive probability to each element of $[0, 1]$, we would have to assign a probability of 0 to each element, since there are uncountably many elements in $[0, 1]$. However, this would lead to a total probability of 0, which is not valid. Of course, we know that the probability of choosing a specific element or a countable set of elements from $[0, 1]$ is 0, but there then arises the question of how to assign a probability to an uncountable set of elements, such as the Cantor set. We can work in a slightly different viewpoint.

Definition 1.2. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty set called the sample space, $\mathcal{F}$ is the collection of all subsets of Ω ($\sim 2^\Omega$), and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a function satisfying $\mathbb{P}(\Omega) = 1$ and countable additivity, that is, for any countable collection of pairwise disjoint sets $A_1, A_2, \dots \in \mathcal{F}$,

we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \quad (1.4)$$

Using this definition, we look to define a $\mathbb{P}$ on subsets of $[0, 1]$ that satisfies the above properties, along with the property that $\mathbb{P}([a, b]) = b - a$. Thus, we have $\mathbb{P}([0, 1]) = 1$. If we have two disjoint intervals $I_1, I_2 \subseteq [0, 1]$, then $\mathbb{P}(I_1 \cup I_2) = \mathbb{P}(I_1) + \mathbb{P}(I_2)$. This is a natural property to expect from a probability measure. What about a general set $S \subseteq [0, 1]$? One can observe that if we have a sequence of intervals $I_1, I_2, \dots$ such that $S \subseteq \bigcup_{k=1}^{\infty} I_k$, then we would have $\mathbb{P}(S) \leq \sum_{k=1}^{\infty} \mathbb{P}(I_k)$. This is because the probability of S should be less than or equal to the probability of any set that contains S . Thus, we can define

$$\mathbb{P}_*(S) = \inf \left\{ \sum_{k=1}^{\infty} |I_k| \mid S \subseteq \bigcup_{k=1}^{\infty} I_k, I_k \subseteq [0, 1] \right\}. \quad (1.5)$$

We have not verified that this is a valid definition of a probability measure, hence the subscript $*$. Via this, let us try computing $\mathbb{P}_*(\mathbb{Q} \cap [0, 1])$. Enumerate the rationals in $[0, 1]$ as $\{q_1, q_2, \dots\}$. For any $\epsilon > 0$, we can choose intervals $I_k = [q_k - \epsilon/2^{k+1}, q_k + \epsilon/2^{k+1}] \cap [0, 1]$ for each $k \in \mathbb{N}$. Note that $\bigcup_{k=1}^{\infty} I_k$ contains all the rationals in $[0, 1]$, and $\sum_{k=1}^{\infty} |I_k| \leq \sum_{k=1}^{\infty} \epsilon/2^k = \epsilon$. Since ϵ was arbitrary, we have $\mathbb{P}_*(\mathbb{Q} \cap [0, 1]) = 0$. This is consistent with our intuition that the probability of choosing a rational number from $[0, 1]$ is 0. So, $\mathbb{P}_*$ seems to be a reasonable definition of a probability measure on $([0, 1], 2^{[0, 1]})$. However, we will show later that there exists a set $\mathcal{A} \subseteq [0, 1]$ such that $\mathbb{P}_*(\mathcal{A}) = \mathbb{P}_*([0, 1] \setminus \mathcal{A}) = 1$, contradicting the property that $\mathbb{P}_*([0, 1]) = 1$. We state the following theorem without proof.

Theorem 1.3. *There is no probability measure $\mathbb{P}$ on $([0, 1], 2^{[0, 1]})$ such that $\mathbb{P}([a, b]) = b - a$ for all $[a, b] \subseteq [0, 1]$.*

If we remove the requirement that the probability measure of a subinterval $[a, b]$ is equal to its length, then we can define a probability measure on $([0, 1], 2^{[0, 1]})$. For example, we can define $\mathbb{P}(S) = 1$ if $1/2 \in S$ and $\mathbb{P}(S) = 0$ otherwise. This is a valid probability measure, but it is not very useful. Keeping the requirement that the probability measure of a subinterval $[a, b]$ is equal to its length, we can instead restrict the collection of subsets of $[0, 1]$ on which we define the probability measure.

Definition 1.4. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the sample space,
- $\mathcal{F}$ is a collection of subsets of Ω satisfying the following properties:
 - $\Omega \in \mathcal{F}$,
 - if $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$,
 - if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a function satisfying:
 - $\mathbb{P}(\Omega) = 1$,
 - if $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Such an $\mathcal{F}$ is termed a σ -algebra on Ω , and such a $\mathbb{P}$ is termed a *probability measure* on $(\Omega, \mathcal{F})$.

1.1 σ -algebras

Example 1.5. Given Ω , trivial possible choices for $\mathcal{F}$ are $\{\emptyset, \Omega\}$ and 2^Ω . A more interesting example is the following: suppose Ω is uncountable. Then $\mathcal{F} = \{A \subseteq \Omega \mid A \text{ is countable or } \Omega \setminus A \text{ is countable}\}$ is a σ -algebra on Ω .

Let $S \subseteq 2^\Omega$ be a collection of subsets of Ω . We claim that there exists a notion of the smallest σ -algebra containing S . This is called the σ -algebra generated by S , and is denoted by $\sigma(S)$.

Lemma 1.6. *For any collection $S \subseteq 2^\Omega$, there exists a unique smallest σ -algebra containing S satisfying*

$$\sigma(S) = \bigcap \{ \mathcal{F} \subseteq 2^\Omega \mid \mathcal{F} \text{ is a } \sigma\text{-algebra and } S \subseteq \mathcal{F} \}. \quad (1.6)$$

Proof. It is easy to check that $\sigma(S)$ is indeed a σ -algebra containing S . If $\mathcal{F}$ is any σ -algebra containing S , then $\sigma(S) \subseteq \mathcal{F}$. Thus, $\sigma(S)$ is the unique smallest σ -algebra containing S . ■

Definition 1.7. Let (X, d) be a metric space (or a topological space), and let $\mathcal{S} \subseteq 2^X$ be the collection of all open sets in X . Then, the σ -algebra $\sigma(\mathcal{S})$ is called the *Borel σ -algebra* on X , and its elements are called *Borel sets*.

We will show that there exists a probability measure on $([0, 1], \mathcal{B})$, where $\mathcal{B}$ is the Borel σ -algebra on $[0, 1]$, which extends the notion of length of intervals to Borel sets.

Let us return to the problem of indefinite random walks on the line. Let $\Omega = \{\omega = (\omega_1, \omega_2, \dots) \mid \omega_i \in \{0, 1\}\}$, where $\omega_i = 0$ corresponds to a step to the left and $\omega_i = 1$ corresponds to a step to the right. If we let $A = \{0, 1\}$, then $\Omega = A^\mathbb{N}$; consider the discrete topology on A , and the product topology on Ω . If we let

$$\mathcal{S} = \{U_1 \times U_2 \times \dots \times U_k \times A \times A \times \dots \mid U_i = \{0\} \text{ or } \{1\}\} \quad (1.7)$$

then one can show that the σ -algebra generated by $\mathcal{S}$ is the Borel σ -algebra on Ω . We can set $\mathcal{F} = \sigma(\mathcal{S})$, and define a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ by setting $\mathbb{P}(U_1 \times U_2 \times \dots \times U_k \times A \times A \times \dots) = 2^{-k}$ for all $U_i = \{0\}$ or $\{1\}$.

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Example 1.8. Let (X, d) be a separable metric space (that is, there exists a countable dense subset). The Borel σ -algebra on X is then

$$\mathcal{B}(X) = \sigma(\{B(p, r)\}_{p \in X, r > 0}). \quad (1.8)$$

Note that the usual containment is $\sigma(\{B(p, r)\}_{p \in X, r > 0}) \subseteq \mathcal{B}(X)$ for a metric space X . So, it suffices to show that if $V \in \tau$, the topology on X , then $V \in \sigma(\{B(p, r)\}_{p \in X, r > 0})$. Let Q be a countable dense subset of X . Then, for any $V \in \tau$, we have

$$V = \bigcup_{n=1}^{\infty} B(q_n, r_n), \quad (1.9)$$

where $q_n \in Q \cap V$ and $r_n > 0$ (where r_n is the largest possible radius such that $B(q_n, r_n) \subseteq V$). Since $\sigma(\{B(p, r)\}_{p \in X, r > 0})$ is a σ -algebra, we have $V \in \sigma(\{B(p, r)\}_{p \in X, r > 0})$. Thus, we have $\mathcal{B}(X) = \sigma(\{B(p, r)\}_{p \in X, r > 0})$.

Example 1.9. Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space.

- $\mathcal{F}$ is closed under countable unions, countable intersections, differences, and symmetric differences. Countable unions and intersections follow from the definition of a σ -algebra. If $A, B \in \mathcal{F}$, then $A \setminus B = A \cap (\Omega \setminus B) \in \mathcal{F}$. Finally, $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{F}$.
- Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A_1, A_2, \dots \in \mathcal{F}$. Define

$$\limsup_{n \rightarrow \infty} A_n := \{x \in \Omega \mid x \text{ belongs to infinitely many } A_n\} \quad (1.10)$$

and

$$\liminf_{n \rightarrow \infty} A_n := \{x \in \Omega \mid x \text{ belongs to all but finitely many } A_n\}. \quad (1.11)$$

Then, both the $\limsup$ and $\liminf$ belong in $\mathcal{F}$. To see this, simply note that

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k, \quad \liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k. \quad (1.12)$$

- One may show that $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(\bigcup A_n) \leq \sum \mathbb{P}(A_n)$ for any countable collection of sets $A_1, A_2, \dots \in \mathcal{F}$.
- For a nested sequence of sets $A_1 \subseteq A_2 \subseteq \dots$ in $\mathcal{F}$, let $A = \bigcup_{n=1}^{\infty} A_n$ (this is usually denoted by $A_n \uparrow A$). Then, $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$. To this end, let $B_n = A_n \setminus A_{n-1}$ for $n > 1$, and $B_1 = A_1$. Then $A = \bigcup_{n=1}^{\infty} B_n$. Thus,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{P}(B_n) = \lim_{N \rightarrow \infty} \sum_{n=2}^N \mathbb{P}(A_n) - \mathbb{P}(A_{n-1}) + \mathbb{P}(A_1) = \lim_{N \rightarrow \infty} \mathbb{P}(A_N). \quad (1.13)$$

1.1.1 Other Systems of Sets

The following are other interesting collection of subsets of Ω besides the Borel σ -algebra.

1. π -system: a collection of subsets $S \subseteq \Omega$ such that $A, B \in S$ implies $A \cap B \in S$.
2. λ -system: a collection Λ of subsets of Ω such that $\Omega \in \Lambda$, if $A, B \in \Lambda$ with $A \subseteq B$, then $B \setminus A \in \Lambda$, and if $A_1, A_2, \dots \in \Lambda$, then $\bigcup_{n=1}^{\infty} A_n \in \Lambda$.
3. algebra: a collection $\mathcal{A}$ of subsets of Ω such that $\Omega \in \mathcal{A}$, if $A \in \mathcal{A}$, then $\Omega \setminus A \in \mathcal{A}$, and if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

We had defined $\omega(S)$, the σ -algebra generated by a collection of subsets $S \subseteq 2^\Omega$. We can similarly define $\pi(S)$, the π -system generated by S , $\lambda(S)$, the λ -system generated by S , and $\mathcal{A}(S)$, the algebra generated by S .

Theorem 1.10 (*Sierpiński-Dynkin π - λ Theorem*). *Let $S, \mathcal{F} \subseteq 2^\Omega$ be collections of subsets of Ω . Then,*

1. $\mathcal{F}$ is a σ -algebra if and only if $\mathcal{F}$ is both a π -system and a λ -system.
2. If S is a π -system, then $\lambda(S) = \sigma(S)$.

Proof. 1. If $\mathcal{F}$ is a σ -algebra, then it is trivially both a π -system and a λ -system. Conversely, if $\mathcal{F}$ is both a π -system and a λ -system, we need to verify the axioms of a σ -algebra. We have $\Omega \in \mathcal{F}$ since $\mathcal{F}$ is a λ -system. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$ since $\mathcal{F}$ is a λ -system. Finally, if $A_1, A_2, \dots \in \mathcal{F}$, then we can write $B_n = \bigcup_{i=1}^n A_i \in \mathcal{F}$ since $A_i^c \in \mathcal{F}$ and $B_n^c = \bigcap_{i=1}^n A_i^c \in \mathcal{F}$ since $\mathcal{F}$ is a π -system. Then, $B_n \uparrow B = \bigcup_{i=1}^{\infty} A_i$, and since $\mathcal{F}$ is a λ -system, we have $B \in \mathcal{F}$. Thus, $\mathcal{F}$ is a σ -algebra.

2. Let S be a π -system. Since $\lambda(S) \subseteq \sigma(S)$, it suffices to show that $\lambda(S)$ is a π -system. Fix $A \in S$, and define $\mathcal{C}_A = \{B \in \lambda(S) \mid A \cap B \in \lambda(S)\}$. Note that $S \subseteq \mathcal{C}_A$ because S is a π -system. We show that $\mathcal{C}_A$ is a λ -system. We have $\Omega \in \mathcal{C}_A$ since $A \cap \Omega = A \in \lambda(S)$. If $B_1, B_2 \in \mathcal{C}_A$ with $B_1 \subseteq B_2$, then $A \cap B_1 \subseteq A \cap B_2$, and since $A \cap B_2 \in \lambda(S)$, we have $(A \cap B_2) \setminus (A \cap B_1) = A \cap (B_2 \setminus B_1) \in \lambda(S)$, and thus $B_2 \setminus B_1 \in \mathcal{C}_A$. Finally, if $B_1, B_2, \dots$ are in $\mathcal{C}_A$, then $A \cap B_n \in \lambda(S)$ for all n , and thus $\bigcup (A \cap B_n) = A \cap (\bigcup B_n) \in \lambda(S)$, and hence $\bigcup B_n \in \mathcal{C}_A$. Thus, $\mathcal{C}_A$ is a λ -system. Since $\lambda(S)$ is the smallest λ -system containing S , we have $\lambda(S) = \mathcal{C}_A$, and hence for any $B, C \in S$, we have $B, C \in S$ implies $B \cap C = A \cap C \in S$. Thus, $\lambda(S)$ is a π -system. ■

Lemma 1.11. *Let S be a π -system, and let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$. Then, $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(S)$.*

Proof. Let $\mathcal{C} = \{B \in \sigma(S) \mid \mathbb{P}(B) = \mathbb{Q}(B)\}$. Then, $S \subseteq \mathcal{C}$, and we show that $\mathcal{C}$ is a λ -system. We have $\Omega \in \mathcal{C}$ since $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$. If $B_1, B_2 \in \mathcal{C}$ with $B_1 \subseteq B_2$, then $\mathbb{P}(B_2 \setminus B_1) = \mathbb{P}(B_2) - \mathbb{P}(B_1) = \mathbb{Q}(B_2) - \mathbb{Q}(B_1) = \mathbb{Q}(B_2 \setminus B_1)$, and hence $B_2 \setminus B_1 \in \mathcal{C}$. Finally, if $B_1, B_2, \dots$ are in $\mathcal{C}$, then $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$, and since $\mathbb{Q}(A) = \lim_{n \rightarrow \infty} \mathbb{Q}(A_n)$, we have $\mathbb{P}(A) = \mathbb{Q}(A)$. ■

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Example 1.12. We wish to obtain an example of S which is *not* a π -system, and two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$, but $\mathbb{P}(A) \neq \mathbb{Q}(A)$ for some $A \in \sigma(S)$. Let $\Omega = \{1, 2, 3, 4\}$, and let $S = \{\{1, 2\}, \{2, 3\}\}$. Then, $\sigma(S) = 2^\Omega$. Define $\mathbb{P}$ and $\mathbb{Q}$ on $(\Omega, \sigma(S))$ by

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = 1/4, \quad (1.14)$$

and

$$\mathbb{Q}(\{1\}) = \mathbb{Q}(\{3\}) = 1/6, \quad \mathbb{Q}(\{2\}) = \mathbb{Q}(\{4\}) = 1/3. \quad (1.15)$$

Then, $\mathbb{P}(\{1, 2\}) = \mathbb{Q}(\{1, 2\}) = 1/2$ and $\mathbb{P}(\{2, 3\}) = \mathbb{Q}(\{2, 3\}) = 1/2$, but $\mathbb{P}(\{3\}) = 1/4 \neq 1/6 = \mathbb{Q}(\{3\})$.

Example 1.13. Let $\Omega \neq \emptyset$, and A_λ be a collection of subsets of Ω indexed by $\lambda \in \Lambda$. Consider the σ -algebra $\sigma(\{A_\lambda\}_{\lambda \in \Lambda})$, and pick $B \in \sigma(\{A_\lambda\}_{\lambda \in \Lambda})$. We will show that there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$. Let $\mathcal{C} = \{B \in \sigma(\{A_\lambda\}_{\lambda \in \Lambda}) \mid B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty) \text{ for some countable subcollection } \{A_{\lambda_n}\}_{n=1}^\infty\}$. Then, $\mathcal{C}$ is a σ -algebra containing $\{A_\lambda\}_{\lambda \in \Lambda}$, and hence $\sigma(\{A_\lambda\}_{\lambda \in \Lambda}) \subseteq \mathcal{C}$. Thus, $B \in \mathcal{C}$, and hence there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$.

1.2 The Lebesgue Measure

Theorem 1.14. *There exists a unique Borel measure λ on $[0, 1]$ such that $\lambda(I) = |I|$ for all intervals $I \subseteq [0, 1]$.*

Proof. For the π -system $S = \{(a, b) \mid 0 \leq a < b \leq 1\}$, we know that $\sigma(S)$ is a Borel σ -algebra on $[0, 1]$. If λ and μ are two Borel measures on $[0, 1]$ such that $\lambda(I) = \mu(I) = |I|$ for all intervals $I \subseteq [0, 1]$, then by the previous lemma, we have $\lambda(A) = \mu(A)$ for all $A \in \sigma(S)$. Thus, the Borel measure is unique.

To show the existence of such a measure, we will define a pseudomeasure λ_* that satisfies the interval property, and then show that it is a measure on the Borel σ -algebra. For any $E \subseteq [0, 1]$, define

$$\lambda_*(E) = \inf \left\{ \sum_{i=1}^\infty |I_i| \mid I_i \text{ are intervals and } E \subseteq \bigcup_{i=1}^\infty I_i \right\} \quad (1.16)$$

Infer that $0 \leq \lambda_*(E) \leq 1$ for all $E \subseteq [0, 1]$. Moreover, let A and B be two sets in $[0, 1]$. Then, for any $\varepsilon > 0$, we have

$$\lambda_*(A \cup B) \leq \sum_{k=1}^\infty |I_k| + \sum_{k=1}^\infty |J_k| \leq \lambda_*(A) + \varepsilon + \lambda_*(B) + \varepsilon \implies \lambda_*(A \cup B) \leq \lambda_*(A) + \lambda_*(B). \quad (1.17)$$

Finally, let $(a, b) \subseteq [0, 1]$ be an interval. Then, $\lambda_*((a, b)) = b - a$. To see this, suppose $(a, b) \subseteq \bigcup_{k=1}^\infty I_k$ for some intervals I_k . Now consider $[a + \varepsilon, b]$ for some $\varepsilon > 0$ such that $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^\infty I_k$. This is a compact subinterval, so we can extract a finite subcover $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^N I_k$. Then,

$$b - a - \varepsilon = |[a + \varepsilon, b]| \leq \sum_{k=1}^N |I_k| \leq \sum_{k=1}^\infty |I_k| \implies b - a \leq \lambda_*((a, b)). \quad (1.18)$$

Now, since $(a, b] \subseteq (a, b + \varepsilon)$, a similar line of reasoning shows that $\lambda_*((a, b]) \leq b - a$. Thus, we have $\lambda_*((a, b]) = b - a$. The property of finite subadditivity also implies the property of countable subadditivity (show this). We now construct the σ -algebra $\mathcal{F}$ on which λ_* is a measure. We pick

$$\mathcal{F} = \{A \subseteq [0, 1] \mid \lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c) \text{ for all } E \subseteq [0, 1]\}. \quad (1.19)$$

We show that $\mathcal{F}$ is a σ -algebra containing the Borel σ -algebra, and that λ_* is a measure on $\mathcal{F}$. It is left as a minor exercise to show that $\mathcal{F}$ is closed under intersections and complements (making $\mathcal{F}$ a π -system). We show $\mathcal{F}$ is a λ -system. Let $A, B \in \mathcal{F}$. Then $B \setminus A = B \cap A^c$ is in $\mathcal{F}$. Then, for any $E \subseteq [0, 1]$, we have

$$\lambda_*(E) = \lambda_*(E \cap A_n) + \lambda_*(E \cap A_n^c) \quad (1.20)$$

$$\geq \lambda_*(E \cap A_n) + \lambda_*(E \cap A^c), \quad (1.21)$$

because $A^c \subseteq A_n^c$. Since $A_n \uparrow A$, we have $E \cap A_n \uparrow E \cap A$, and hence

$$\lambda_*(E \cap A_n) \uparrow \lambda_*(E \cap A).$$

Passing to the limit gives

$$\lambda_*(E) \geq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

On the other hand, subadditivity of the outer measure gives

$$\lambda_*(E) \leq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

Therefore,

$$\lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c),$$

so $A \in \mathcal{F}$. ■

July 30th.

Note that $\mathcal{F} \subsetneq 2^{[0,1]}$; that is, there exist subsets of $[0, 1]$ that are not in $\mathcal{F}$. We shall assume that $\lambda_*(E) = \lambda_*(E + x)$ for all $E \subseteq [0, 1]$ and $x \in \mathbb{R}$ such that $E + x \subseteq [0, 1]$. This is a reasonable assumption, since we expect the Lebesgue measure to be translation invariant (one can prove this). Moreover, instead of looking at $[0, 1]$ we shall look at $([0, 1], \oplus)$, where $\oplus$ is addition modulo 1. Now suppose we manage to break $[0, 1]$ into countably many disjoint pieces $A_1, A_2, \dots$ such that $\bigcup_{n=1}^{\infty} A_n = [0, 1]$, and $A_i = A_1 \oplus r_i$ for some $r_i \in [0, 1]$. Then, we would have $\lambda_*(A_i) = \lambda_*(A_1)$ for all i , and hence $\lambda_*([0, 1]) = \sum_{n=1}^{\infty} \lambda_*(A_n) = \sum_{n=1}^{\infty} \lambda_*(A_1)$. This is a contradiction, since $\lambda_*([0, 1]) = 1$ and $\sum_{n=1}^{\infty} \lambda_*(A_1)$ is either 0 or ∞ . Thus, we have shown that there exists a set $A \subseteq [0, 1]$ such that $A \notin \mathcal{F}$.

To construct these A_i 's, simply look at the equivalence relation $\sim$ on $[0, 1]$ defined by $x \sim y$ if and only if $x - y \in \mathbb{Q}$. Then, we can pick a representative from each equivalence class to form a set A_1 . Then, we can define $A_i = A_1 \oplus r_i$ for some $r_i \in [0, 1]$ such that $\{r_i\}_{i=1}^{\infty}$ is a complete set of representatives of $\mathbb{Q} \cap [0, 1]$. This gives us the desired decomposition of $[0, 1]$ into disjoint pieces.

1.2.1 Carathéodory's Extension Theorem

August 4th.

Suppose Ω is non-empty, and $\mathcal{A}$ is an *algebra*; that is, $\mathcal{A}$ is a collection of subsets of Ω such that $\Omega \in \mathcal{A}$, and $\mathcal{A}$ is closed under complements and finite unions. Further say that μ is a *pre-measure* on $\mathcal{A}$; that is, $\mu : \mathcal{A} \rightarrow [0, 1]$ satisfies $\mu(\emptyset) = 0$, and if $A_1, A_2, \dots \in \mathcal{A}$ are pairwise disjoint, then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$. Then, we can extend μ to a measure on the σ -algebra generated by $\mathcal{A}$. This is the content of *Carathéodory's extension theorem*.

Remark 1.15. 1. Note that since $\mathcal{A}$ is a π -system, such an extension must be unique.

2. For a π -system S , a pre-measure on S gives a pre-measure on $\mathcal{A}(S)$, the algebra generated by S , since every set $B \in \mathcal{A}(S)$ can be written as a finite disjoint union of sets in S .

We now wish to assign a 'correct' setting to talk about infinite coin tosses (or simple random walks on $\mathbb{Z}$). Let $\Omega = \{0, 1\}^{\mathbb{N}}$, and we define

$$S = \{\{\mathbf{x} \mid x_1 = a_1, x_2 = a_2, \dots, x_n = a_n\} \mid n \in \mathbb{N}, a_i \in \{0, 1\}\} \cup \{\emptyset\} \quad (1.22)$$

and let $\mathcal{F} = \sigma(S)$. Verify that S is a π -system; we can set

$$\mu(\{\mathbf{x} \mid x_1 = a_1, x_2 = a_2, \dots, x_n = a_n\}) = \prod_{i=1}^n p^{a_i} (1-p)^{1-a_i}. \quad (1.23)$$

For now, assume that μ is a pre-measure on S . Then there'll exist a unique probability measure on $(\Omega, \sigma(S))$ that extends μ .

1.3 Measurable Maps and Random Variables

Note that a space $(\Omega, \mathcal{F})$ is called a *measurable space*.

Definition 1.16. A map $T : (\Omega, \mathcal{F}) \rightarrow (\tilde{\Omega}, \tilde{\mathcal{F}})$ between two measurable spaces is termed a *measurable function* if $T^{-1}(A) \in \mathcal{F}$ for all $A \in \tilde{\mathcal{F}}$. When $(\tilde{\Omega}, \tilde{\mathcal{F}}) = (\mathbb{R}, \mathcal{B})$, we call T a *random variable*.

In most situations, X is usually a topological space and $\mathcal{B}(X)$ is the Borel σ -algebra on X . For the case of $X = \mathbb{R}$, a random variable can be interpreted as a meaningful quantity which can be studied.

Proposition 1.17. $T : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ is measurable if and only if $T^{-1}(U) \in \mathcal{F}$ whenever U is open in $\mathbb{R}$.

Proof. One direction is clear; for the converse direction, let $\tilde{\mathcal{F}} = \{W \subseteq \mathbb{R} \mid T^{-1}(W) \in \mathcal{F}\}$. It is relatively easy to show that $\tilde{\mathcal{F}}$ is a σ -algebra on $\mathbb{R}$. Thus, since $\mathcal{B} \subseteq \tilde{\mathcal{F}}$, we have $T^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{B}$. ■

Note that the image space being in $\mathbb{R}$ did not play a role in the above proof; for any image space $\tilde{\Omega}$, we can define $\tilde{\mathcal{F}} = \{W \subseteq \tilde{\Omega} \mid T^{-1}(W) \in \mathcal{F}\}$, and show that $\tilde{\mathcal{F}}$ is a σ -algebra on $\tilde{\Omega}$. Such a $\tilde{\mathcal{F}}$ is called the *pushforward σ -algebra* of $\mathcal{F}$ under T , and is the largest σ -algebra on $\tilde{\Omega}$ such that T is measurable. One can think in the converse way, where we are given a map $T : \Omega \rightarrow (\tilde{\Omega}, \tilde{\mathcal{F}})$; to make such a T measurable, one can simply equip Ω with the powerset σ -algebra 2^Ω , and then T is automatically measurable. Alternatively, we can define $\mathcal{F} = \{T^{-1}(A) \mid A \in \tilde{\mathcal{F}}\}$, and then T is measurable. This is called the *pullback σ -algebra* of $\tilde{\mathcal{F}}$ under T , and is the smallest σ -algebra on Ω such that T is measurable.

The most non-trivial yet basic example of a random variable is the *indicator function* of a set $A \in \mathcal{F}$, which is defined as $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$, and 0 otherwise. It is easy to check that $\mathbf{1}_A$ is a random variable.

Lemma 1.18. If $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ are random variables, then $X + Y$ is also a random variable. More generally, if $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous, then $f(X, Y)$ is a random variable.

August 7th.

Example 1.19. Consider the infinite simple random walk from before; for $a \in \mathbb{Z}$, define $\tau_a = \inf\{n > 0 \mid S_n = a\}$, the first hitting time of a . Then we claim that τ_a is measurable and, hence, a random variable. To see this, note that $\tau_a^{-1}((n, \infty)) = \{\omega \in \Omega \mid \tau_a(\omega) > n\} = \{\omega \in \Omega \mid S_1(\omega) \neq a, S_2(\omega) \neq a, \dots, S_n(\omega) \neq a\} = \bigcap_{k=1}^n \{\omega \in \Omega \mid S_k(\omega) \neq a\}$, which is in $\mathcal{F}$ since S_k is measurable for all k . Thus, τ_a is measurable.

Definition 1.20. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $T : (\Omega, \mathcal{F}) \rightarrow (\tilde{\Omega}, \tilde{\mathcal{F}})$ be a measurable map. Then, the *pushforward measure* of $\mathbb{P}$ under T is the measure $\tilde{\mathbb{P}}$ on $(\tilde{\Omega}, \tilde{\mathcal{F}})$ defined by $\tilde{\mathbb{P}}(A) = \mathbb{P}(T^{-1}(A))$ for all $A \in \tilde{\mathcal{F}}$.

Indeed, one can show that the pushforward measure $\tilde{\mathbb{P}}$ is a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{F}})$.

Definition 1.21. Given a Borel probability measure μ on $(\mathbb{R}, \mathcal{B})$, we define the *cumulative distribution function* of μ to be the function $F_\mu : \mathbb{R} \rightarrow [0, 1]$ defined by $F_\mu(x) = \mu((-\infty, x])$ for all $x \in \mathbb{R}$.

Proposition 1.22. *With μ as above, the following hold true.*

1. F_μ is non-decreasing.
2. $\lim_{x \rightarrow \infty} F_\mu(x) = 1$ and $\lim_{x \rightarrow -\infty} F_\mu(x) = 0$.
3. F_μ is right-continuous.

Proof. 1. If $x_1 \leq x_2$, then $(-\infty, x_1] \subseteq (-\infty, x_2]$, and hence $F_\mu(x_1) = \mu((-\infty, x_1]) \leq \mu((-\infty, x_2]) = F_\mu(x_2)$.

2. Note that $\mathbb{R} = \bigcup_{n=1}^{\infty} (-\infty, n]$; defining $A_k = (-\infty, k]$, we have $A_1 \subseteq A_2 \subseteq \dots$ and $A_k \uparrow \mathbb{R}$. Since μ is a measure, $\mu(A_k) \uparrow \mu(\mathbb{R}) = 1$, and hence $\lim_{x \rightarrow \infty} F_\mu(x) = 1$. Similarly, $\lim_{x \rightarrow -\infty} F_\mu(x) = 0$.

3. Let x_n be a decreasing sequence of numbers such that $x_n \downarrow x$. Then, $(-\infty, x_1] \supseteq (-\infty, x_2] \supseteq \dots$ and $(-\infty, x_n] \downarrow (-\infty, x]$. Since μ is a measure, $\mu((-\infty, x_n]) \downarrow \mu((-\infty, x])$, and hence $F_\mu(x_n) \downarrow F_\mu(x)$. Thus, F_μ is right-continuous. ■

Theorem 1.23. *Suppose $F : \mathbb{R} \rightarrow [0, 1]$ is a non-decreasing, right-continuous function such that $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. Then, there exists a unique Borel probability measure μ on $(\mathbb{R}, \mathcal{B})$ such that $F = F_\mu$.*

Thus, there is a one-to-one correspondence between Borel probability measures on $(\mathbb{R}, \mathcal{B})$ and cumulative distribution functions. In general, it is easier to study the cumulative distribution function of a probability measure than the measure itself.

1.3.1 The Pushforward Measure

August 11th.

In many situations, we can model a random experiment in several ways. Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}$, and λ the Lebesgue measure. Define the random variables $X_1 = \mathbf{1}_{[0, 1/2]}$ and $X_2 = \mathbf{1}_{[0, 1/4] \cup [1/2, 3/4]}$. Consider $\Gamma = (X_1, X_2) : \Omega \rightarrow \mathbb{R}^2$, and let $\mu = \lambda \circ \Gamma^{-1}$ be the pushforward measure of λ under Γ . The support of μ is the set $\Gamma(\Omega) = \{(1, 0), (0, 1), (1, 1), (0, 0)\}$. The measure at each of these points is given by $\mu(\{(1, 0)\}) = \lambda(\Gamma^{-1}(\{(1, 0)\})) = \lambda([0, 1/4]) = 1/4$, $\mu(\{(0, 1)\}) = \lambda(\Gamma^{-1}(\{(0, 1)\})) = \lambda([1/2, 3/4]) = 1/4$, $\mu(\{(1, 1)\}) = \lambda(\Gamma^{-1}(\{(1, 1)\})) = \lambda([1/4, 1/2]) = 1/4$, and $\mu(\{(0, 0)\}) = \lambda(\Gamma^{-1}(\{(0, 0)\})) = \lambda([3/4, 1]) = 1/4$. Thus, we have a probability measure on the support of μ . That is, we have modeled tossing a fair coin twice, without explicitly defining the sample space as $\{0, 1\}^2$. The probability space does not matter in most contexts; the random variables and their distributions are what matter.

Naturally, we can ask further questions. Whether the following models can exist: tossing a coin n times, and tossing a coin countably many times. The first question is easy to answer; we can define the k -th random variable X_k exactly as

$$X_k = \mathbf{1}_{[0, \frac{1}{2^k}] \cup [\frac{2}{2^k}, \frac{3}{2^k}] \cup \dots \cup [\frac{2^k-2}{2^k}, \frac{2^k-1}{2^k}]}. \quad (1.24)$$

We leave it as an exercise to verify that $(X_1, X_2, \dots, X_n)$ has the same distribution as tossing a fair coin n times.

Theorem 1.24. *Given any probability measure μ on $(\mathbb{R}, \mathcal{B})$, there exists a measurable map $T : ([0, 1], \mathcal{B}([0, 1]), \lambda) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ such that $\mu = \lambda \circ T^{-1}$. That is, the pushforward measure of λ under T is μ .*

Before we proceed with this theorem, we show the following proposition.

Proposition 1.25. *For any Borel probability measures μ and ν on $(\mathbb{R}, \mathcal{B})$, if $F_\mu = F_\nu$, then $\mu = \nu$. That is, the cumulative distribution function uniquely determines the probability measure.*

Proof. Let μ and ν be the required measures such that $F_\mu = F_\nu$, or $\mu((-\infty, x]) = \nu((-\infty, x])$ for all $x \in \mathbb{R}$. Let $S = \{(-\infty, x] \mid x \in \mathbb{R}\}$, which is a π -system. Note that $\sigma(S) = \mathcal{B}$; thus, by the Sierpiński-Dynkin π - λ theorem, we have $\mu(A) = \nu(A)$ for all $A \in \mathcal{B}$. Hence, $\mu = \nu$. ■

We now prove **Theorem 1.23** and **Theorem 1.24** together.

Proof of theorems. Suppose $F : \mathbb{R} \rightarrow [0, 1]$ be a non-decreasing, right-continuous function such that $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. We will show that there exists a measurable map $T : ([0, 1], \mathcal{B}([0, 1]), \lambda) \rightarrow (\mathbb{R}, \mathcal{B})$ such that $F_{\lambda \circ T^{-1}} = F$; note that this implies the second theorem by the previous proposition.

For such an F , we can define a generalized inverse T defined as $T(x) = \inf\{p \in \mathbb{R} \mid F(p) \geq x\}$. For any $x \in [0, 1]$, the set $\{p \in \mathbb{R} \mid F(p) \geq x\}$ is non-empty (since $\lim_{p \rightarrow \infty} F(p) = 1$) and has infimum, so T is well-defined. Moreover, T is non-decreasing and takes values in $\mathbb{R}$. To show measurability of T , note that for any $a \in \mathbb{R}$, we have

$$\{x \in [0, 1] \mid T(x) > a\} = \{x \in [0, 1] \mid F(a) < x\} = (F(a), 1] \cap [0, 1], \quad (1.25)$$

which is a Borel set in $[0, 1]$. Thus, T is measurable. We now verify that $F_{\lambda \circ T^{-1}} = F$; For any $x \in \mathbb{R}$, we have

$$\begin{aligned} (\lambda \circ T^{-1})((-\infty, x]) &= \lambda(T^{-1}((-\infty, x])) = \lambda(\{u \in [0, 1] \mid T(u) \leq x\}) \\ &= \lambda(\{u \in [0, 1] \mid \inf\{p \in \mathbb{R} \mid F(p) \geq u\} \leq x\}) = \lambda(\{u \in [0, 1] \mid F(x) \geq u\}) \\ &= \lambda([0, F(x)]) = F(x). \end{aligned} \quad (1.26)$$

Thus, $F_{\lambda \circ T^{-1}} = F$, establishing the existence of the desired measure. ■

1.4 Probability Measures on $\mathbb{R}$

August 14th.

We first wish to generalize the notion of the integral of a function with respect to the Lebesgue measure. A function $\varphi : [a, b] \rightarrow \mathbb{R}$ is said to be a *simple function* if it is of the form $\varphi = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ for some $n \in \mathbb{N}$, $a_i \in \mathbb{R}$, and $A_i \in \mathcal{B}$. That is, φ takes finitely many values, and the pre-image of each value is a Borel set. For a simple function φ , we define the integral of φ to be

$$\int_{[a,b]} \varphi(x) dx = \sum_{i=1}^n a_i \lambda(A_i). \quad (1.27)$$

In general, a function $g : [a, b] \rightarrow \mathbb{R}$ is said to be *Borel measurable* if $g^{-1}(A) \in \mathcal{B}$ for all Borel sets $A \subseteq \mathbb{R}$. Note that any simple function is Borel measurable. The integral of a non-negative Borel measurable function $g : [a, b] \rightarrow \mathbb{R}$ is then defined as

$$\int_{[a,b]} g(x) dx = \sup \left\{ \int_{[a,b]} \varphi(x) dx \mid 0 \leq \varphi \leq g, \varphi \text{ is simple} \right\}. \quad (1.28)$$

One can infer that $\int(f + \alpha g) = \int f + \alpha \int g$, $g \leq f \implies \int g \leq \int f$, and $\int_{I_1 \sqcup I_2} g = \int_{I_1} g + \int_{I_2} g$ for disjoint intervals I_1 and I_2 , and non-negative Borel measurable functions f and g . We can extend the definition of the integral to all Borel measurable functions g by defining $\int g = \int g^+ - \int g^-$, where $g^+ = \max\{g, 0\}$ and $g^- = \max\{-g, 0\}$. Thus, we are fit to define the following.

Definition 1.26. A probability measure $\mu \in \mathcal{P}(\mathbb{R})$ is said to have a *density* f if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a non-negative Borel measurable function such that $\int_{\mathbb{R}} f(x) dx = 1$ and $F_\mu(x) = \int_{-\infty}^x f(t) dt$ for all $x \in \mathbb{R}$.

Suppose $\mu \in \mathcal{P}(\mathbb{R})$ has a density f . Then, for any Borel set $A \subseteq \mathbb{R}$, we have

$$\mu(A) = \int_A f(x) dx. \quad (1.29)$$

Note that not every probability measure has a density; for example, the measure $\mu = \delta_0$, indicator of the set $\{0\}$, does not have a density. In fact, if there exists $p \in \mathbb{R}$ with $\mu(\{p\}) > 0$, then μ does not have a

density. This is because if μ has a density f , then $\mu(\{p\}) = \int_{\{p\}} f(x) dx = 0$, which is a contradiction. Thus, any probability measure with a density must be non-atomic; that is, it cannot assign positive measure to any singleton set. However, the converse is not true; there exist non-atomic probability measures that do not have a density. For example, consider the Cantor measure μ on $[0, 1]$, which is non-atomic, but does not have a density.

We now wish to define a metric on the space of probability measures on $\mathbb{R}$.

1. For any $\mu, \nu \in \mathcal{P}(\mathbb{R})$, we define the *total variation distance* between μ and ν to be

$$D_1(\mu, \nu) = d_{TV}(\mu, \nu) := \sup_{A \in \mathcal{B}} |\mu(A) - \nu(A)|. \quad (1.30)$$

2. The CDFs may be utilized to define

$$D_2(\mu, \nu) := \sup_{x \in \mathbb{R}} |F_\mu(x) - F_\nu(x)|. \quad (1.31)$$

Both of these metrics are fine as they are, but fail in being ‘nice’ with respect to convergence of measures; the sequence of measures $\{\mu_n = \delta_{1/n}\}_{n=1}^\infty$ fails to converge to δ_0 in either of these metrics, even though we expect it to converge intuitively.

August 18th.

Hereforth, we define the *Levy distance* on $\mathcal{P}(\mathbb{R})$ as

$$d(\mu, \nu) := \inf\{\epsilon > 0 \mid F_\mu(t + \epsilon) + \epsilon \geq F_\nu(t), F_\nu(t + \epsilon) + \epsilon \geq F_\mu(t) \text{ for all } t \in \mathbb{R}\}. \quad (1.32)$$

In fact, this can be generalized to a metric on Borel probability measures on any metric space $(X, \mathcal{D})$ as follows (this is termed the *Levy-Prokhorov metric*).

$$d_{LP}(\mu, \nu) := \inf\{\epsilon > 0 \mid \mu(A + \epsilon) + \epsilon \geq \nu(A), \nu(A + \epsilon) + \epsilon \geq \mu(A) \text{ for all closed } A \subseteq X\} \quad (1.33)$$

where the ϵ -fattening of A is defined as

$$A + \epsilon := \{x \in X \mid \text{there exists } y \in A \text{ such that } \mathcal{D}(x, y) < \epsilon\}. \quad (1.34)$$

Note that these two metrics are not the same on $\mathcal{P}(\mathbb{R})$, but are equivalent. Also, via the Levy distance, we can show $d(\delta_{1/n}, \delta_0) = 1/n$, showing that $\delta_{1/n} \rightarrow \delta_0$ in this metric.

August 25th.

Theorem 1.27. Suppose $\mu, \mu_1, \mu_2, \dots \in \mathcal{P}(\mathbb{R})$ are Borel probability measures. Then $d(\mu_n, \mu) \xrightarrow{n \rightarrow \infty} 0$ if and only if $F_{\mu_n}(x) \xrightarrow{n \rightarrow \infty} F_\mu(x)$ for all continuity points $x \in \mathbb{R}$ of F_μ .

Proof. We first show the forward implication. Suppose $d(\mu_n, \mu) \rightarrow 0$. Then, for any $\epsilon > 0$, there exists $N_\epsilon \in \mathbb{N}$ such that $d(\mu_n, \mu) < \epsilon$ for all $n \geq N_\epsilon$. Now fix $\epsilon > 0$ and $x \in \mathbb{R}$. Then, for all $n \geq N_\epsilon$, we have $F_\mu(x + \epsilon) + \epsilon \geq F_{\mu_n}(x)$. We can replace the right hand side with the limit superior to get

$$F_\mu(x + \epsilon) + \epsilon \geq \limsup_{n \rightarrow \infty} F_{\mu_n}(x) \quad \text{for all } \epsilon > 0. \quad (1.35)$$

This implies that $F_\mu(x) \geq \limsup_{n \rightarrow \infty} F_{\mu_n}(x)$. Similarly, for all $n \geq N_\epsilon$, we have $F_{\mu_n}(x) + \epsilon \geq F_\mu(x - \epsilon)$. By a similar argument, we have $\liminf_{n \rightarrow \infty} F_{\mu_n}(x) \geq F_\mu(x)$ if x is a continuity point of F_μ . Thus, we have shown that $\limsup_{n \rightarrow \infty} F_{\mu_n}(x) \leq F_\mu(x) \leq \liminf_{n \rightarrow \infty} F_{\mu_n}(x)$ or $F_{\mu_n}(x) \rightarrow F_\mu(x)$ for all continuity points.

For the converse, if $\mathcal{C}$ denotes the set of continuity points of F_μ , then $\mathcal{C}$ is cocountable and hence dense in $\mathbb{R}$. Now fix $\epsilon \in (0, 1)$. Let $A, B \in \mathbb{R}$ be numbers such that $F_\mu(A) < \epsilon$ and $F_\mu(B) \geq 1 - \epsilon$. Pick points $\{x_i\}_{i=0}^m$ in $[A, B] \cap \mathcal{C}$ such that $A = x_0 < x_1 < \dots < x_m = B$ and $x_{i+1} - x_i < \epsilon$ for all i . By the choice of the x_i 's, we have $F_{\mu_n}(x_i) \rightarrow F_\mu(x_i)$ for all i , and so there exists $N_\epsilon \in \mathbb{N}$ such that $|F_{\mu_n}(x_i) - F_\mu(x_i)| < \epsilon$ for all $n \geq N_\epsilon$ and all i . Now fix $t \in [A, B]$; so $t \in [x_i, x_{i+1}]$ for some i . Then, for all $n \geq N_\epsilon$, we have

$$F_{\mu_n}(t + \epsilon) \geq F_{\mu_n}(x_{i+1}) \geq F_\mu(x_{i+1}) - \epsilon \geq F_\mu(t) - \epsilon \implies F_{\mu_n}(t + \epsilon) + \epsilon \geq F_\mu(t) \quad (1.36)$$

for all $n \geq N_\varepsilon$ and $t \in [A, B]$. If $t > B$, then both $F_{\mu_n}(t)$ and $F_\mu(t)$ are at least $1 - 2\varepsilon$, and hence $F_{\mu_n}(t + \varepsilon) + \varepsilon \geq F_\mu(t)$. If $t < A$, then both $F_{\mu_n}(t)$ and $F_\mu(t)$ are at most 2ε , and hence $F_{\mu_n}(t + \varepsilon) + \varepsilon \geq F_\mu(t)$. Thus, we have shown that for all $\varepsilon > 0$, there exists $N_\varepsilon \in \mathbb{N}$ such that for all $n \geq N_\varepsilon$, we have

$$F_{\mu_n}(t + \varepsilon) + \varepsilon \geq F_\mu(t) \quad \text{for all } t \in \mathbb{R} \implies d(\mu_n, \mu) \leq 2\varepsilon. \quad (1.37)$$

■

1.5 Expectation

Expectation, in our new setting, corresponds to integration in a probability space. To this end, we start with simple functions.

Definition 1.28. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A measurable function $S : \Omega \rightarrow \mathbb{R}$ is called a *simple random variable* if it takes finitely many values; that is, $S(\Omega) = \{s_1, \dots, s_n\}$ for some $n \in \mathbb{N}$.

In this case, we can write $S = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ for some $c_i \in \mathbb{R}$ and $A_i \in \mathcal{F}$. This representation need not be unique; however, note that any such simple function can be written in a unique way as

$$S = \sum_{i=1}^n \beta_i \mathbf{1}_{B_i} \quad (1.38)$$

where $\beta_i \in S(\Omega)$ are unequal and B_i 's are measurable sets that are mutually disjoint.

Definition 1.29. The *expectation of a simple random variable* $S = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ is defined as

$$\mathbb{E}[S] := \sum_{i=1}^n c_i \mathbb{P}(A_i). \quad (1.39)$$

Since this representation is not unique, one can verify that the expectation is well-defined by using the unique representation of S as $\sum_{i=1}^n \beta_i \mathbf{1}_{B_i}$. Also, if S and T are simple random variables, then $S + T$ is also a simple random variable, and $\mathbb{E}[S + T] = \mathbb{E}[S] + \mathbb{E}[T]$. Similarly, for any $\alpha \in \mathbb{R}$, we have $\mathbb{E}[\alpha S] = \alpha \mathbb{E}[S]$. If $S \geq 0$, then $\mathbb{E}[S] \geq 0$. We can now extend the definition of expectation to all non-negative random variables, and then to all random variables.

Definition 1.30. The *expectation of a non-negative random variable* $X : \Omega \rightarrow [0, \infty)$ is defined as

$$\mathbb{E}[X] := \sup\{\mathbb{E}[S] \mid 0 \leq S \leq X, S \text{ is a simple random variable}\}. \quad (1.40)$$

If X is a general random variable, we can write $X = X^+ - X^-$, where $X^+ = \max\{X, 0\}$ and $X^- = \max\{-X, 0\}$. If $\mathbb{E}[X^+] < \infty$ or $\mathbb{E}[X^-] < \infty$, then we define the *expectation of (the) random variable* X as

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-]. \quad (1.41)$$

Proposition 1.31. Let X and Y be non-negative random variables. Then, the following hold true.

1. $\mathbb{E}[X] \geq 0$.
2. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, if at least one of $\mathbb{E}[X]$ or $\mathbb{E}[Y]$ is finite.

Proof. The first part is clear from the definition since $S \equiv 0$ is a simple random variable such that $0 \leq S \leq X$, so $\mathbb{E}[X] \geq \mathbb{E}[S] = 0$. For the second part, let S and T be simple random variables such that $0 \leq S \leq X$ and $0 \leq T \leq Y$. Then, $S + T$ is a simple random variable such that $0 \leq S + T \leq X + Y$, and hence $\mathbb{E}[X + Y] \geq \mathbb{E}[S + T] = \mathbb{E}[S] + \mathbb{E}[T]$. Taking supremum over all such S and T , we get $\mathbb{E}[X + Y] \geq \mathbb{E}[X] + \mathbb{E}[Y]$. For the reverse inequality, let U be a simple random variable such that $0 \leq U \leq X + Y$. Then, we can write $U = U_1 + U_2$ where $U_1 = U \mathbf{1}_{\{X > 0\}}$ and $U_2 = U \mathbf{1}_{\{Y > 0\}}$. Then,

$0 \leq U_1 \leq X$ and $0 \leq U_2 \leq Y$, and hence $\mathbb{E}[U] = \mathbb{E}[U_1] + \mathbb{E}[U_2] \leq \mathbb{E}[X] + \mathbb{E}[Y]$. Taking supremum over all such U , we get $\mathbb{E}[X + Y] \leq \mathbb{E}[X] + \mathbb{E}[Y]$. Thus, we have equality. ■

1.5.1 Results of Convergence

August 28th.

We also have the monotone converge theorem, of which we state a basic version first.

Theorem 1.32. *Suppose $X \geq 0$ is a random variable, and $\{\varphi_n\}_{n=1}^\infty$ is a sequence of simple random variables such that $0 \leq \varphi_n \uparrow X$. Then, $\mathbb{E}[\varphi_n] \uparrow \mathbb{E}[X]$.*

Proof. Fix $\varepsilon > 0$. By definition, there exists a simple function S such that $0 \leq S \leq X$ and $\mathbb{E}[S] \geq \mathbb{E}[X] - \varepsilon$. For a fixed $\omega \in \Omega$, there exists a natural $N_\omega \in \mathbb{N}$ such that $\varphi_n(\omega) \geq S(\omega) - \varepsilon$ for all $n \geq N_\omega$. Now define $A_n = \{\omega \in \Omega \mid \varphi_n(\omega) \geq S(\omega) - \varepsilon\}$; since φ_n 's are increasing functions, we have $A_1 \subseteq A_2 \subseteq \dots \subseteq A_n \subseteq A_{n+1} \subseteq \dots$, and $\bigcup_{n=1}^\infty A_n = \Omega$. Thus, given $\delta > 0$, we can find a natural $N_\delta \equiv N$ such that $\mathbb{P}(A_{N_\delta}) \geq 1 - \delta$. Now, for all $n \geq N$, we have

$$\mathbb{E}[\varphi_n] = \mathbb{E}[\varphi_n \mathbf{1}_{A_N}] + \mathbb{E}[\varphi_n \mathbf{1}_{A_N^c}]. \quad (1.42)$$

But note that on such an A_N , we have $\varphi_n \geq S - \varepsilon$, and hence $\mathbb{E}[\varphi_n \mathbf{1}_{A_N}] \geq \mathbb{E}[(S - \varepsilon) \mathbf{1}_{A_N}] = \mathbb{E}[S \mathbf{1}_{A_N}] - \varepsilon \mathbb{P}(A_N)$. If $M = \sup_{\omega \in \Omega} S(\omega)$, then $\mathbb{E}[\varphi_n \mathbf{1}_{A_N^c}] \geq 0$ and $\mathbb{E}[S \mathbf{1}_{A_N}] = \mathbb{E}[S] - \mathbb{E}[S \mathbf{1}_{A_N^c}] \geq \mathbb{E}[S] - M \mathbb{P}(A_N^c)$. Thus, we have

$$\mathbb{E}[\varphi_n] \geq \mathbb{E}[S] - M \mathbb{P}(A_N^c) - \varepsilon \mathbb{P}(A_N) \geq \mathbb{E}[S] - M \delta - \varepsilon. \quad (1.43)$$

Choosing $\delta = \varepsilon/M$, we have $\mathbb{E}[\varphi_n] \geq \mathbb{E}[S] - 2\varepsilon \geq \mathbb{E}[X] - 3\varepsilon$. Since $\varepsilon > 0$ was arbitrary, we have $\lim_{n \rightarrow \infty} \mathbb{E}[\varphi_n] = \mathbb{E}[X]$. ■

Theorem 1.33 (The monotone convergence theorem). *Suppose $X \geq 0$ is a random variable, and $\{X_n\}_{n=1}^\infty$ is a sequence of non-negative random variables such that $0 \leq X_n \uparrow X$. Then, $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.*

Lemma 1.34 (Fatou's lemma). *Suppose $\{X_n\}_{n=1}^\infty$ is a sequence of non-negative random variables. Then,*

$$\mathbb{E}[\liminf_{n \rightarrow \infty} X_n] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n]. \quad (1.44)$$

Proof. Define $Y_n = \inf_{k \geq n} X_k$ for all $n \in \mathbb{N}$. Then, $\{Y_n\}_{n=1}^\infty$ is a sequence of non-negative random variables such that $Y_1 \leq Y_2 \leq \dots$ and $Y_n \uparrow \liminf_{n \rightarrow \infty} X_n$. By the monotone convergence theorem, we have

$$\mathbb{E}[\liminf_{n \rightarrow \infty} X_n] = \lim_{n \rightarrow \infty} \mathbb{E}[Y_n] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n], \quad (1.45)$$

where the last inequality follows from the fact that $Y_n \leq X_n$ for all n . ■

September 1st.

Theorem 1.35 (The dominated convergence theorem). *Suppose $\{X_n\}_{n=1}^\infty$ is a sequence of random variables such that $X_n \rightarrow X$ almost surely, and there exists a non-negative random variable Y such that $|X_n| \leq Y$ for all n , and $\mathbb{E}[Y] < \infty$. Then, $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$.*

Proof. Define $Z_n = 2Y - |X_n - X|$. Since $Z_n \geq 0$, we can apply Fatou's lemma to get

$$\mathbb{E}[\liminf_{n \rightarrow \infty} Z_n] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[Z_n] \implies 2\mathbb{E}[Y] - \mathbb{E}[\limsup_{n \rightarrow \infty} |X_n - X|] \leq 2\mathbb{E}[Y] - \limsup_{n \rightarrow \infty} \mathbb{E}[|X_n - X|], \quad (1.46)$$

which implies that $\mathbb{E}[\limsup_{n \rightarrow \infty} |X_n - X|] \geq \limsup_{n \rightarrow \infty} \mathbb{E}[|X_n - X|]$. Now, since $X_n \rightarrow X$ almost surely, we have $\limsup_{n \rightarrow \infty} |X_n - X| = 0$ almost surely, and hence $\mathbb{E}[\limsup_{n \rightarrow \infty} |X_n - X|] = 0$. Thus, we have $\limsup_{n \rightarrow \infty} \mathbb{E}[|X_n - X|] = 0$, which implies that $\mathbb{E}[|X_n - X|] \rightarrow 0$, and hence $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$. ■

Chapter 2

MORE ON RANDOM VARIABLES

2.1 Independence

September 15th.

Definition 2.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space.

1. We say sub- σ -algebras $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n$ of $\mathcal{F}$ are independent if $\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2) \dots \mathbb{P}(A_n)$ for all $A_i \in \mathcal{G}_i, i = 1, 2, \dots, n$.
2. We say that the random variables $X_1, X_2, \dots, X_n$ are independent if the sub- σ -algebras $\sigma(X_1), \sigma(X_2), \dots, \sigma(X_n)$ are independent.
3. We say that the events $A_1, A_2, \dots, A_n$ are independent if the sub- σ -algebras $\sigma(A_1), \sigma(A_2), \dots, \sigma(A_n)$ are independent.

Equivalent condition for independence of events is that

$$\mathbb{P}(A_1^\pm \cap A_2^\pm \cap \dots \cap A_n^\pm) = \mathbb{P}(A_1^\pm)\mathbb{P}(A_2^\pm) \dots \mathbb{P}(A_n^\pm), \quad (2.1)$$

where $A_i^+ = A_i$ and $A_i^- = A_i^c$ for $i = 1, 2, \dots, n$.

Definition 2.2. We say that a collection of countably many random variables $X_1, X_2, \dots$ on $(\Omega, \mathcal{F}, \mathbb{P})$ are independent if every finite subcollection of them are independent.

Definition 2.3. Suppose $X_1, X_2, \dots$ are random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and define $\mathcal{F}_n = \sigma(X_n, X_{n+1}, \dots)$ for $n = 1, 2, \dots$. The *tail σ -algebra* of the sequence $X_1, X_2, \dots$ is defined as

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \mathcal{F}_n \subseteq \sigma(X_1, X_2, \dots) = \mathcal{F}. \quad (2.2)$$

An event $A \in \mathcal{T}$ is an event whose occurrence is determined by the variables $X_n, X_{n+1}, \dots$ for every n . Thus, A does not depend on any fixed finite number of the variables at the beginning of the sequence. For example, the event

$$A = \{X_n \text{ occurs infinitely often}\} \quad (2.3)$$

belongs to $\mathcal{T}$, since for every m it can be described using only $X_m, X_{m+1}, \dots$

If $X_1, X_2, \dots$ are independent and $A \in \mathcal{T}$, then A is independent of $\sigma(X_1, \dots, X_n)$ for every n , since A is measurable with respect to $\sigma(X_{n+1}, X_{n+2}, \dots)$. Therefore A is independent of

$$\sigma(X_1, X_2, \dots) = \sigma\left(\bigcup_{n=1}^{\infty} \sigma(X_1, \dots, X_n)\right). \quad (2.4)$$

In particular, A is independent of itself, and hence

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2. \quad (2.5)$$

Thus $\mathbb{P}(A) \in \{0, 1\}$. This is known as *Kolmogorov's zero-one law*, that for independent random variables, any tail event has probability 0 or 1. Without independence, a tail event may retain information about the variables in every finite initial segment, and this argument does not apply.

2.1.1 Percolation

September 18th.

We first discuss (Bernoulli) percolation on $\mathbb{Z}^2 \subseteq \mathbb{R}^2$. To each edge e , we associate $X_e \sim \text{Ber}(p)$ for some $p \in [0, 1]$, independently for each edge. If $X_e = 1$, we retain the edge e , and if $X_e = 0$, we delete the edge e . We then have a random subgraph of $\mathbb{Z}^2$, which is a random subset of edges of $\mathbb{Z}^2$. The present or retained edges are termed open, and the absent or deleted edges are termed closed. We ask if there exists an unbounded connected component of open edges; let E_p be the event that there exists an unbounded connected component of open edges. Since there are only countably many X_e 's, we can order them in a spiral like manner starting from the origin, and let $X_1, X_2, \dots$ be the corresponding sequence of random variables. Then E_p is a tail event here, and hence by Kolmogorov's zero-one law, $\mathbb{P}(E_p) \in \{0, 1\}$. Moreover, $\mathbb{P}(E_0) = 0$ and $\mathbb{P}(E_1) = 1$, and for $0 < p < q < 1$, we have $\mathbb{P}(E_p) \leq \mathbb{P}(E_q)$ by a coupling argument. Therefore, there exists a critical probability $p_c \in [0, 1]$ such that $\mathbb{P}(E_p) = 0$ for $p < p_c$ and $\mathbb{P}(E_p) = 1$ for $p > p_c$.

Let $\{0 \leftrightarrow \infty\}$ denote the event that there exists an unbounded connected component of open edges containing the origin. Define $\theta_p := \mathbb{P}(0 \leftrightarrow \infty)$. Note that this is not a tail event, but $\theta_p > 0$ if and only if $\mathbb{P}(E_p) = 1$. Also, θ_p is non-decreasing in p .

Lemma 2.4. *For small p , $\mathbb{P}(E_p) = 0$. For $p \approx 1$, $\mathbb{P}(E_p) = 1$.*

Proof. Firstly, note that the event $\{0 \leftrightarrow \infty\}$ is contained in the event that there exists a self avoiding path of open edges starting from the origin and ending at manhattan distance N from the origin, for some $N \in \mathbb{N}$. This event is further contained in the event that there exists a self avoiding path of open edges starting from the origin with length at least N . Call the first event A_N and the second event B_N . Then $\{0 \leftrightarrow \infty\} \subseteq A_N \subseteq B_N$, and hence $\mathbb{P}(0 \leftrightarrow \infty) \leq \mathbb{P}(A_N) \leq \mathbb{P}(B_N)$. If C_N is the event that there exists a self avoiding path of open edges starting from the origin of length exactly N , then $\mathbb{P}(C_N) \leq 4 \cdot 3^{N-1} p^N \leq (4p)^N$, since there are at most $4 \cdot 3^{N-1}$ self avoiding paths of length N starting from the origin, and each such path is open with probability p^N . Moreover, note that $B_N = \bigcup_{n=N}^{\infty} C_n$, and hence $\mathbb{P}(B_N) \leq \sum_{n=N}^{\infty} (4p)^n$. Therefore, if $p < 1/4$, we have $\mathbb{P}(B_N) = \frac{(4p)^N}{1-4p} \rightarrow 0$ as $N \rightarrow \infty$, and hence $\mathbb{P}(0 \leftrightarrow \infty) = 0$ for $p < 1/4$. So, for small p , specifically for $p < 1/4$, we have $\theta_p = 0$ and hence $\mathbb{P}(E_p) = 0$.

For the second statement, note that the $\text{Ber}(p)$ percolation model on $\mathbb{Z}^2$ induces a $\text{Ber}(1-p)$ percolation model on the dual lattice $(\mathbb{Z}^2)^*$. Thus, the event $\{0 \nleftrightarrow \infty\}$ is exactly the event that there exists a loop of open edges in the dual lattice surrounding the origin. If C_N^* is the event that there exists a loop of open edges in the dual lattice surrounding the origin with length exactly N , then this event is contained in the event that there exists a self avoiding path of open edges in the dual lattice starting at $\{(1, 0), (2, 0), \dots, (n, 0)\}$ of length exactly N . Therefore, $\mathbb{P}(C_N^*) \leq n(4(1-p))^N$, and hence $\mathbb{P}(\{0 \nleftrightarrow \infty\}) \leq \sum_{N=1}^{\infty} n(4(1-p))^N$. Therefore, if $p > 3/4$, we have $\mathbb{P}(\{0 \nleftrightarrow \infty\}) = 0$, and hence $\theta_p = 1$ and $\mathbb{P}(E_p) = 1$ for $p > 3/4$. ■

Lemma 2.5 (*Borel-Cantelli lemma*). *Let $A_1, A_2, \dots$ be a sequence of events in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*
2. *If $A_1, A_2, \dots$ are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

We prove the second statement.

Proof. Let $B_n = \bigcup_{j=n}^{\infty} A_j$, so that $B_1 \supseteq B_2 \supseteq \dots$ and $\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}(\bigcap_{n=1}^{\infty} B_n)$. If we prove that $\mathbb{P}(B_n) = 1$ for all n , then we are done. Now, $\mathbb{P}(B_n) = 1 - \mathbb{P}(B_n^c) = 1 - \mathbb{P}(\bigcap_{j=n}^{\infty} A_j^c)$. Since $A_1, A_2, \dots$ are

independent, we have

$$\mathbb{P}\left(\bigcap_{j=n}^m A_j^c\right) = \prod_{j=n}^m \mathbb{P}(A_j^c) = \prod_{j=n}^m (1 - \mathbb{P}(A_j)) \leq \prod_{j=n}^m e^{-\mathbb{P}(A_j)} = e^{-\sum_{j=n}^m \mathbb{P}(A_j)}. \quad (2.6)$$

Taking the limit as $m \rightarrow \infty$, we get $\mathbb{P}(\bigcap_{j=n}^{\infty} A_j^c) \leq e^{-\sum_{j=n}^{\infty} \mathbb{P}(A_j)} = 0$, since $\sum_{j=n}^{\infty} \mathbb{P}(A_j) = \infty$. Therefore, $\mathbb{P}(B_n) = 1$ for all n , and hence $\mathbb{P}(A_n \text{ i.o.}) = 1$. ■

2.2 Moments

For a random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R}$, recall that the *mean* of X was simply its expectation $\mathbb{E}[X]$, if it exists. Else, the mean is undefined. For higher moments, the *p-th moment* of X is defined as $\mathbb{E}[|X|^p]$, if it exists. The *variance* of X is defined as $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

In many cases, for a given X , we may not know its exact distribution, but we may know its moments. In such cases, we can use the moments to bound the probabilities of certain events. For example, if X is a non-negative random variable with finite mean $\mathbb{E}[X]$, then by Markov's inequality, for any $t > 0$, we have

$$\mathbb{P}(X > t) \leq \frac{\mathbb{E}[X]}{t}. \quad (2.7)$$

In particular, if $\mathbb{E}[X] = 0$, then $\mathbb{P}(X \geq t) = 0$ for all $t > 0$, and hence $X = 0$ almost surely. If X also happens to have a finite second moment $\mathbb{E}[X^2]$, then we can replace X with $|X - \mathbb{E}[X]|$ in the above inequality to get Chebyshev's inequality, which states that for any $t > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| > t) \leq \frac{\text{Var}(X)}{t^2}. \quad (2.8)$$

In fact, if $\mathbb{E}[|X - \mathbb{E}[X]|^n] < \infty$, then

$$\mathbb{P}(|X - \mu| > t) \leq \frac{\mathbb{E}[|X - \mu|^n]}{t^n}. \quad (2.9)$$

The *exponential moment* of X is defined as $\mathbb{E}[e^{\lambda x}]$ for $\lambda \neq 0$, if it exists. If $\mathbb{E}[e^{\lambda x}] < \infty$ for some $\lambda > 0$, then by Markov's inequality, we have

$$\mathbb{P}(X > t) = \mathbb{P}(e^{\lambda X} > e^{\lambda t}) \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t}}. \quad (2.10)$$

Note that if the p -th moment of X exists, then the q -th moment of X also exists for all $0 < q < p$; this holds since if we write $|X|^p = |X|^p \mathbf{1}_{\{|X| \leq 1\}} + |X|^p \mathbf{1}_{\{|X| > 1\}}$, and note that $\mathbb{E}[|X|^p]$ is finite if and only if $\mathbb{E}[|X|^p \mathbf{1}_{\{|X| > 1\}}]$ is finite, then

$$\mathbb{E}[|X|^q] = \mathbb{E}[|X|^q \mathbf{1}_{\{|X| \leq 1\}}] + \mathbb{E}[|X|^q \mathbf{1}_{\{|X| > 1\}}] \leq 1 + \mathbb{E}[|X|^p \mathbf{1}_{\{|X| > 1\}}] < \infty. \quad (2.11)$$

2.3 Convergence of Random Variables

September 22nd.

Throughout, we shall assume that $(X_n)_{n \geq 1}$ is a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and X is another random variable on the same probability space. We shall discuss various notions of convergence of the sequence (X_n) to X .

Definition 2.6. X_n converges to X as *almost sure convergence*, written $X_n \xrightarrow{\text{a.s.}} X$, if there exists a set $E \in \mathcal{F}$ with $\mathbb{P}(E) = 0$ such that

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \quad \text{for all } \omega \in \Omega \setminus E. \quad (2.12)$$

Definition 2.7. X_n converges to X as *convergence in probability*, written $X_n \xrightarrow{P} X$, if

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \epsilon) = 0 \quad \text{for all } \epsilon > 0. \quad (2.13)$$

Almost sure convergence is the stronger notion—

Proposition 2.8. If $X_n \xrightarrow{\text{a.s.}} X$, then $X_n \xrightarrow{P} X$.

Proof. Let $\epsilon > 0$. Since $X_n \xrightarrow{\text{a.s.}} X$, there exists a set $E \in \mathcal{F}$ with $\mathbb{P}(E) = 0$ such that $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$ for all $\omega \in \Omega \setminus E$. Therefore, for each $\omega \in \Omega \setminus E$, there exists $N_\epsilon(\omega) \in \mathbb{N}$ such that $|X_n(\omega) - X(\omega)| \leq \epsilon$ for all $n \geq N_\epsilon(\omega)$. Now define

$$A_k := \{\omega \in \Omega \mid |X_n(\omega) - X(\omega)| \leq \epsilon, \text{ for all } n \geq k\}. \quad (2.14)$$

Then $(A_k)_{k \geq 1}$ is an increasing sequence of sets such that $\emptyset \subseteq A_k \uparrow \Omega \setminus E$; for any $(1 - \delta) > 0$, there exists a natural $N \in \mathbb{N}$ such that $\mathbb{P}(A_N) \geq 1 - \delta$. Then, we must have

$$\{|X_n - X| > \epsilon\} \subseteq \Omega \setminus A_N \implies 0 \leq \mathbb{P}(|X_n - X| > \epsilon) \leq \mathbb{P}(\Omega \setminus A_N) \leq \delta. \quad (2.15)$$

Letting $\delta \rightarrow 0^+$, we get $\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$; since $\epsilon > 0$ was arbitrary, we have $X_n \xrightarrow{P} X$. ■

However, the converse does not hold. Consider the sequence of random variables $(X_n)_{n \geq 1}$ on $([0, 1], \mathcal{B}([0, 1]), \lambda)$ defined as

$$X_{2^k+j}(x) = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k})}(x), \quad k = 0, 1, 2, \dots, \quad j = 0, 1, \dots, 2^k - 1. \quad (2.16)$$

Let $X = 0$. For $n = 2^k + j$, we have

$$\mathbb{P}(|X_n - X| > \epsilon) = 2^{-k} \longrightarrow 0 \quad (0 < \epsilon < 1), \quad (2.17)$$

and the probability is zero for $\epsilon \geq 1$. Hence $X_n \xrightarrow{P} 0$. On the other hand, every $x \in [0, 1)$ belongs to exactly one interval at each level k , so $X_n(x) = 1$ infinitely often, while $X_n(x) = 0$ infinitely often. Thus X_n does not converge almost surely to 0 (indeed, it fails to converge for every $x \in [0, 1)$).

Lemma 2.9. Suppose $X_n \xrightarrow{P} X$ and, additionally,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \epsilon) < \infty \quad \text{for all } \epsilon > 0. \quad (2.18)$$

Then $X_n \xrightarrow{\text{a.s.}} X$.

Proof. Let $\epsilon > 0$. By the Borel-Cantelli lemma, we have $\mathbb{P}(|X_n - X| > \epsilon \text{ i.o.}) = 0$. Let $E_\epsilon = \{|X_n - X| > \epsilon \text{ i.o.}\}$, so that $\mathbb{P}(E_\epsilon) = 0$. Then for all $\omega \in \Omega \setminus E_\epsilon$, there exists $N_\epsilon(\omega) \in \mathbb{N}$ such that $|X_n(\omega) - X(\omega)| \leq \epsilon$ for all $n \geq N_\epsilon(\omega)$. Now define

$$E = \bigcup_{m=1}^{\infty} E_{1/m}. \quad (2.19)$$

Then $\mathbb{P}(E) = 0$, and for all $\omega \in \Omega \setminus E$, we have $|X_n(\omega) - X(\omega)| \leq 1/m$ for all n sufficiently large, for every $m = 1, 2, \dots$. Therefore, $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$ for all $\omega \in \Omega \setminus E$, and hence $X_n \xrightarrow{\text{a.s.}} X$. ■

Let $(X_n)_{n \geq 1}$ be a sequence of pairwise independent random variables with $\mathbb{E}[X_n] = 0$ and $\text{Var}(X_n) \leq M$ for all n and some constant $M > 0$. Then if $S_n = X_1 + \dots + X_n$, we have $S_n/n \xrightarrow{P} 0$ since

$$\mathbb{P}(|S_n/n| > \epsilon) \leq \frac{\mathbb{E}[|\frac{S_n}{n}|^2]}{\epsilon^2} = \frac{\text{Var}(S_n/n)}{\epsilon^2} = \frac{\text{Var}(S_n)}{n^2 \epsilon^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2 \epsilon^2} \leq \frac{M}{n \epsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (2.20)$$

Let $(X_n)_{n \geq 1}$ be bounded independent random variables with $\mathbb{E}[X_n] = 0$ for all n . Then if $S_n = X_1 + \dots + X_n$, we have $S_n/n \xrightarrow{\text{a.s.}} 0$; to this end, we have

$$\mathbb{P}(|S_n/n| > \epsilon) \leq \frac{\mathbb{E}[|S_n/n|^4]}{\epsilon^4} = \frac{\mathbb{E}[S_n^4]}{n^4 \epsilon^4} \leq \frac{C(n + n^2)M^4}{n^4 \epsilon^4} \lesssim \frac{1}{n^2}. \quad (2.21)$$

Thus,

$$\sum_{n=1}^{\infty} \mathbb{P}(|S_n/n| > \epsilon) < \infty, \quad (2.22)$$

or $|S_n/n| \xrightarrow{\text{a.s.}} 0$ by the previous lemma.

2.4 Second Moment Method

September 25th.

Let $X \geq 0$ be a non-negative random variable. Given just this, note that $\mathbb{P}(X > 0)$ cannot have a lower bound since we could very well define X to be 0 with probability $1 - \epsilon$ and 1 with probability ϵ , for any $\epsilon > 0$. If we specify that $\mathbb{E}[X]$ has to be ‘large’, we still cannot guarantee that $\mathbb{P}(X > 0)$ is large, since we could define X to be 0 with probability $1 - \epsilon$ and $1/\epsilon$ with probability ϵ , for any $1 > \epsilon > 0$.

Suppose we are given $\mu = \mathbb{E}[X]$. Then by Markov’s inequality, we have

$$\mathbb{P}(X > \frac{\mu}{2}) = \mathbb{P}(|X - \mu| \leq \frac{\mu}{2}) + \mathbb{P}(X > \frac{3\mu}{2}) \geq 1 - \mathbb{P}(|X - \mu| > \frac{\mu}{2}) \geq 1 - \frac{4 \text{Var}(X)}{\mu^2} \geq 1 - 4\delta \quad (2.23)$$

when $\text{Var}(X) \leq \delta \mu^2$ for some $\delta > 0$. Thus, if $\text{Var}(X)$ is small compared to $\mathbb{E}[X]^2$, then X is concentrated around its mean, and hence $\mathbb{P}(X > 0)$ is large. Note that the choice of $\mu/2$ is arbitrary; we have the following generalization.

Lemma 2.10 (The second moment method). *Let $X \geq 0$ be a non-negative random variable, and $0 \leq \alpha \leq 1$. Then*

$$\mathbb{P}(X > \alpha \mathbb{E}[X]) \geq (1 - \alpha)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \quad (2.24)$$

In particular,

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \quad (2.25)$$

Proof. Recall the Cauchy-Schwarz inequality, which states that for any random variables X and Y , we have $\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$. Let $Y = \mathbf{1}_{\{X > \alpha \mathbb{E}[X]\}}$. Then $\mathbb{E}[Y^2] = \mathbb{E}[Y] = \mathbb{P}(X > \alpha \mathbb{E}[X])$, and $\mathbb{E}[XY] = \mathbb{E}[X \mathbf{1}_{\{X > \alpha \mathbb{E}[X]\}}] = \mathbb{E}[X] - \mathbb{E}[X \mathbf{1}_{\{X \leq \alpha \mathbb{E}[X]\}}] \geq (1 - \alpha)\mathbb{E}[X]$. Therefore, by Cauchy-Schwarz,

$$\mathbb{P}(X > \alpha \mathbb{E}[X]) = \mathbb{E}[Y^2] \geq \frac{\mathbb{E}[XY]^2}{\mathbb{E}[X^2]} \geq (1 - \alpha)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \quad (2.26)$$

■

We may also prove the Borel-Cantelli lemma using the second moment method, the statement being that if $A_1, A_2, \dots$ are independent events (or simply pairwise independent) with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Proof. Let $B_n = \bigcup_{i=n}^{\infty} A_i$. Then $B_n \downarrow \{A_n \text{ i.o.}\}$. Fix n and define $S_k := \sum_{j=n}^k \mathbf{1}_{A_j}$ for $k \geq n$. Then, with

$$\mathbb{P}(A_i) = \alpha_i,$$

$$\begin{aligned} \mathbb{P}(S_k \geq 1) &= \mathbb{P}(S_k > 0) \geq \frac{\mathbb{E}[S_k]^2}{\mathbb{E}[S_k^2]} = \frac{(\sum_{j=n}^k \mathbb{P}(A_j))^2}{\sum_{j=n}^k \mathbb{P}(A_j) + 2 \sum_{n \leq i < j \leq k} \mathbb{P}(A_i \cap A_j)} \\ &= \frac{(\sum_{j=n}^k \alpha_j)^2}{\sum_{j=n}^k \alpha_j + 2 \sum_{n \leq i < j \leq k} \alpha_i \alpha_j} = \frac{(\sum_{i=n}^k \alpha_i)^2}{(\sum_{i=n}^k \alpha_i) + (\sum_{i=n}^k \alpha_i)^2} \\ &\geq \frac{1}{1 + \frac{1}{\sum_{i=n}^k \alpha_i}}. \end{aligned} \quad (2.27)$$

Letting $k \rightarrow \infty$, we get

$$\mathbb{P}(B_n) = \lim_{k \rightarrow \infty} \mathbb{P}(S_k \geq 1) \geq 1. \quad (2.28)$$

Since this holds for all n , we have $\mathbb{P}(A_n \text{ i.o.}) = 1$. ■

2.4.1 The Coupon Collector Problem

n different books are placed on a bookshelf. Every night, a book is chosen uniformly at random, read, and then returned to the shelf. Let T_n be the number of days until all n books have been read. We want to find $\mathbb{E}[T_n]$. To this end, define $Z_1 = 1$, and for $2 \leq k \leq n$, let Z_k be the number of days after the first $k-1$ books have been read until a new book is read. Then $T_n = Z_1 + Z_2 + \cdots + Z_n$. Note that $Z_2 \sim \text{Geom}(\frac{n-1}{n})$, since there are $n-1$ books that have not been read yet with probability $\frac{n-1}{n}$ of reading a new book, and similarly $Z_k \sim \text{Geom}(\frac{n-k+1}{n})$ for $2 \leq k \leq n$. Therefore, we have

$$\mathbb{E}[T_n] = \sum_{k=1}^n \mathbb{E}[Z_k] = \sum_{k=1}^n \frac{n}{n-k+1} = n \sum_{k=1}^n \frac{1}{k} \sim n \log n. \quad (2.29)$$

Now let θ_n be a sequence of positive numbers such that $\theta_n \rightarrow \infty$ as $n \rightarrow \infty$. We claim that

$$\mathbb{P}(|T_n - n \log n| \geq n\theta_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (2.30)$$

To show this, set $t \in \mathbb{N}$, and define

$$S_t := \sum_{j=1}^n \mathbf{1}_{B_j}, \quad (2.31)$$

where B_j is the event that book j has not been picked in the first t days. Then S_t is the number of books that have not been picked in the first t days, and hence $\{T_n > t\} = \{S_t \geq 1\} = \{S_t > 0\}$. Computing moments of S_t , while later setting $t = n \log n + n\theta_n$, we have

$$\mathbb{E}[S_t] = \sum_{j=1}^n \mathbb{P}(B_j) = n \left(1 - \frac{1}{n}\right)^t \geq n e^{-(\frac{1}{n} + \frac{1}{n^2})t} = e^{\theta_n - \frac{1}{n}(\log n - \theta_n)}. \quad (2.32)$$

and

$$\begin{aligned} \mathbb{E}[S_t^2] &= \sum_{j=1}^n \mathbb{P}(B_j) + 2 \sum_{1 \leq i < j \leq n} \mathbb{P}(B_i \cap B_j) = n \left(1 - \frac{1}{n}\right)^t + n(n-1) \left(1 - \frac{2}{n}\right)^t \\ &\leq n e^{-\frac{t}{n}} + n^2 e^{-\frac{2t}{n}} = e^{\theta_n} + e^{2\theta_n}. \end{aligned} \quad (2.33)$$

Therefore, by the second moment method, we have

$$\mathbb{P}(T_n > n \log n - n\theta_n) = \mathbb{P}(S_t > 0) \geq \frac{\mathbb{E}[S_t]^2}{\mathbb{E}[S_t^2]} \geq \frac{e^{2\theta_n - \frac{2}{n}(\log n - \theta_n)}}{e^{\theta_n} + e^{2\theta_n}} \implies \mathbb{P}(T_n > n \log n - n\theta_n) \xrightarrow{n \rightarrow \infty} 1. \quad (2.34)$$

On the other hand, setting $t = n \log n + n\theta_n$ and using Markov's inequality, we have

$$\begin{aligned} \mathbb{P}(T_n > n \log n + n\theta_n) &= \mathbb{P}(S_t \geq 1) \leq \mathbb{E}[S_t] \leq n \left(1 - \frac{1}{n}\right)^t \leq n e^{-\frac{1}{n}n(\log n + \theta_n)} = e^{-\theta_n} \\ &\implies \mathbb{P}(T_n > n \log n + n\theta_n) \xrightarrow{n \rightarrow \infty} 0. \end{aligned} \quad (2.35)$$

Therefore, we conclude that

$$\mathbb{P}(|T_n - n \log n| \geq n\theta_n) = \mathbb{P}(T_n > n \log n + n\theta_n) + \mathbb{P}(T_n < n \log n - n\theta_n) \xrightarrow{n \rightarrow \infty} 0. \quad (2.36)$$

Dividing by $n \log n$, and setting $\theta_n = \epsilon \log n$ for any $\epsilon > 0$, we get

$$\frac{T_n}{n \log n} \xrightarrow{\mathbb{P}} 1. \quad (2.37)$$

2.4.2 Random Series

Let $(a_n)_{n \geq 0}$ be a sequence of independent and identically distributed random variables with $a_n \sim \text{Exp}(1)$, which has a density function $e^{-x} \mathbf{1}_{x \geq 0}$. Now define

$$f(t) = \sum_{n=0}^{\infty} a_n t^n \quad \text{for all } t \in \mathbb{R}. \quad (2.38)$$

This is a random power series, and we want to find the radius of convergence, R , of this series. We know that R is related to the coefficients a_n as

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}. \quad (2.39)$$

This limit is a random variable which is measurable with respect to the tail- σ -algebra of the sequence $(a_n)_{n \geq 0}$, and hence by Kolmogorov's zero-one law, there exists a constant $c \in [0, \infty]$ such that $\mathbb{P}(R = c) = 1$. We claim that $c = 1$. First note that if we set $t = 1 + \epsilon$ for some $\epsilon > 0$, then

$$\mathbb{P}(|a_n| > t^n) = e^{-t^n} \leq \frac{1}{(1 + \epsilon)^n} \quad \text{for all } n \text{ sufficiently large.} \quad (2.40)$$

Thus, $\sum_{n=0}^{\infty} \mathbb{P}(|a_n| > t^n) < \infty$, and hence by the Borel-Cantelli lemma, we have $\mathbb{P}(|a_n|^{1/n} > t \text{ i.o.}) = 0$, or $\mathbb{P}(\limsup_{n \rightarrow \infty} |a_n|^{1/n} > t) = 0$. Since this holds for all $t > 1$, we have $\mathbb{P}(\limsup_{n \rightarrow \infty} |a_n|^{1/n} > 1) = 0$. On the other hand, if we set $t = 1 - \epsilon$ for some $\epsilon > 0$, then

$$\sum_{n=0}^{\infty} \mathbb{P}(|a_n| > t^n) = \sum_{n=0}^{\infty} e^{-t^n} = \infty, \quad (2.41)$$

so that by the Borel-Cantelli lemma, we have $\mathbb{P}(|a_n|^{1/n} > t \text{ i.o.}) = 1$, or $\mathbb{P}(\limsup_{n \rightarrow \infty} |a_n|^{1/n} > t) = 1$. Since this holds for all $t < 1$, we have $\mathbb{P}(\limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1) = 0$. Therefore, we conclude that $\mathbb{P}(\limsup_{n \rightarrow \infty} |a_n|^{1/n} = 1) = 1$, and hence $\mathbb{P}(R = 1) = 1$.

2.5 Law of Large Numbers

September 29th.

Theorem 2.11 (The basic weak law of large numbers). *Let $(X_n)_{n \geq 1}$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$ and $\text{Var}(X_1) < \infty$. Then*

$$\frac{S_n}{n} = \frac{X_1 + \cdots + X_n}{n} \xrightarrow{\mathbb{P}} \mu = \mathbb{E}[X_1]. \quad (2.42)$$

Proof. We wish to show that for all $\epsilon > 0$, we have $\mathbb{P}(|S_n/n - \mu| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$. To this end, we first note that by Chebyshev's inequality, we have

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) = \mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu)\right| > n\epsilon\right) \leq \frac{\mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right]}{n^2 \epsilon^2} = \frac{n \text{Var}(X_1)}{n^2 \epsilon^2} = \frac{\text{Var}(X_1)}{n \epsilon^2} \xrightarrow{n \rightarrow \infty} 0. \quad (2.43)$$

■

We can use the weak law of large numbers to prove the Weierstrass approximation theorem, which states that any continuous function on $[0, 1]$ can be uniformly approximated by polynomials. To this end, we make use of the *Bernstein polynomials*, which are defined as follows. For a continuous function $f : [0, 1] \rightarrow \mathbb{R}$, define the n -th Bernstein polynomial of f as

$$Q_{n,f} : [0, 1] \rightarrow \mathbb{R}, \quad Q_{n,f}(p) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} p^k (1-p)^{n-k}. \quad (2.44)$$

Note that if $S_n \sim \text{Bin}(n, p)$, then $\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$, and hence $Q_{n,f}(p) = \mathbb{E}[f(S_n/n)]$. Let $S_n = X_1 + X_2 + \cdots + X_n$, where $X_1, \dots, X_n \sim \text{Ber}(p)$ are independent. Then, we have

$$|Q_{n,f}(p) - f(p)| \leq \mathbb{E} \left[\left| f\left(\frac{S_n}{n}\right) - f(p) \right| \right]. \quad (2.45)$$

Splitting via the indicator function of $\{|S_n/n - p| > \delta\}$ for some $\delta > 0$ we choose later, we have

$$\mathbb{E} \left[\left| f\left(\frac{S_n}{n}\right) - f(p) \right| \right] = \mathbb{E} \left[\left| f\left(\frac{S_n}{n}\right) - f(p) \right| \cdot \mathbf{1}_{|S_n/n - p| \leq \delta} \right] + \mathbb{E} \left[\left| f\left(\frac{S_n}{n}\right) - f(p) \right| \cdot \mathbf{1}_{|S_n/n - p| > \delta} \right]. \quad (2.46)$$

By the uniform continuity of f , given $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/2$ whenever $|x - y| \leq \delta$. Thus, the first of these terms may simply be bounded by $\epsilon/2$. For the second term, we have

$$\mathbb{E} \left[\left| f\left(\frac{S_n}{n}\right) - f(p) \right| \cdot \mathbf{1}_{|S_n/n - p| > \delta} \right] \leq 2 \|f\|_\infty \mathbb{P} \left(\left| \frac{S_n}{n} - p \right| > \delta \right) \leq 2 \|f\|_\infty \frac{\text{Var}(S_n/n)}{\delta^2} = 2 \|f\|_\infty \frac{p(1-p)}{n\delta^2}. \quad (2.47)$$

With $c_f = 2 \|f\|_\infty (1/4)$, the second term is bounded by $c_f/(n\delta^2)$, which goes to 0 as $n \rightarrow \infty$. Therefore, we have $|Q_{n,f}(p) - f(p)| < \epsilon$ for all n sufficiently large, and hence $Q_{n,f} \rightarrow f$ uniformly on $[0, 1]$.

We now prove the weak law of large numbers, where we drop the assumption of finite variance.

Theorem 2.12 (The weak law of large numbers). *Let $(X_n)_{n \geq 1}$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$. Then*

$$\frac{S_n}{n} = \frac{X_1 + \cdots + X_n}{n} \xrightarrow{\mathbb{P}} \mu = \mathbb{E}[X_1]. \quad (2.48)$$

Proof. Note that it suffices to prove the result for $\mu = 0$, since we can always replace X_n with $X_n - \mu$. Since we have the basic weak law of large numbers for random variables with finite variance, we will approximate X_n with a truncated version of itself to reduce to the finite variance case. To this end, for each $n \geq 1$, we can split X_j into two parts as

$$X_j = X_j \mathbf{1}_{|X_j| \leq A_n} + X_j \mathbf{1}_{|X_j| > A_n} = Y_{j,n} + Z_{j,n}, \quad (2.49)$$

where $(A_n)_{n \geq 1}$ is a sequence of positive numbers that diverge properly to infinity, that we choose later. The first of these terms converges pointwise to X_j as $n \rightarrow \infty$, and is bounded uniformly; by the dominated convergence theorem, we have $\mathbb{E}[X_j \mathbf{1}_{|X_j| \leq A_n}] \rightarrow \mathbb{E}[X_j] = 0$ as $n \rightarrow \infty$. The second of these terms converges pointwise to 0 as $n \rightarrow \infty$, and is bounded by $|X_j|$; again by the dominated convergence theorem, we have $\mathbb{E}[X_j \mathbf{1}_{|X_j| > A_n}] \rightarrow 0$ as $n \rightarrow \infty$. Let $\mu_n = \mathbb{E}[Y_{1,n}] = -\mathbb{E}[Z_{1,n}]$. Then $\mu_n \rightarrow 0$ as $n \rightarrow \infty$. Now, each S_n may be split as $S_n = S_n^Y + S_n^Z$, where $S_n^Y = Y_{1,n} + \cdots + Y_{n,n}$ and $S_n^Z = Z_{1,n} + \cdots + Z_{n,n}$. Then, for any $\epsilon > 0$, we have

$$\mathbb{P} \left(\left| \frac{S_n}{n} \right| > \delta \right) = \mathbb{P} \left(\left| \frac{S_n^Y}{n} + \frac{S_n^Z}{n} \right| > \delta \right) \leq \mathbb{P} \left(\left| \frac{S_n^Y}{n} - \mu_n \right| > \frac{\delta}{2} \right) + \mathbb{P} \left(\left| \frac{S_n^Z}{n} + \mu_n \right| > \frac{\delta}{2} \right). \quad (2.50)$$

Via Chebyshev's, the first of these terms is bounded as

$$\mathbb{P} \left(\left| \frac{S_n^Y}{n} - \mu_n \right| > \frac{\delta}{2} \right) \leq \frac{4 \text{Var}(Y_{1,n})}{n\delta^2} \quad (2.51)$$

For $\delta > 0$ fixed, let us estimate this for large enough n such that $|\mu_n| < \frac{\delta}{4}$. For the second term, we have

$$\mathbb{P}\left(\left|\frac{S_n^Z}{n} + \mu_n\right| > \frac{\delta}{2}\right) \leq \mathbb{P}\left(\left|\frac{S_n^Z}{n}\right| > \frac{\delta}{4}\right) \leq \mathbb{P}(\text{at least one of the } Z_{j,n} \text{ is non-zero}) \leq n\mathbb{P}(|X_1| > A_n). \quad (2.52)$$

Combining, we obtain

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \delta\right) \leq \frac{4\text{Var}(Y_{1,n})}{n\delta^2} + n\mathbb{P}(|X_1| > A_n) \leq \frac{4A_n\mathbb{E}[|X_1|]}{n\delta^2} + \frac{n\mathbb{E}[|X_1| \cdot \mathbf{1}_{|X_1| > A_n}]}{A_n}. \quad (2.53)$$

We now wish to choose A_n such that both of these terms go to 0 as $n \rightarrow \infty$. So, given $\epsilon > 0$, choose $A_n = \epsilon n\delta^2/4\mathbb{E}[|X_1|]$. Then, we have

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \delta\right) \leq \epsilon + \frac{4\mathbb{E}[|X_1|]}{\delta^2} \epsilon \mathbb{E}[|X_1| \cdot \mathbf{1}_{|X_1| > A_n}] \xrightarrow{n \rightarrow \infty} 2\epsilon. \quad (2.54)$$

Since $\epsilon > 0$ was arbitrary, we conclude that $\mathbb{P}(|S_n/n| > \delta) \rightarrow 0$ as $n \rightarrow \infty$, and hence $S_n/n \xrightarrow{P} 0$. $\blacksquare$

Example 2.13. Let $X_1, X_2, \dots \sim \text{Exp}(1)$ be independent and identically distributed random variables, and set $M_n := \max\{X_1, \dots, X_n\}$. Then $M_n/\log n \xrightarrow{P} 1$ as $n \rightarrow \infty$. To see this, note that

$$\mathbb{P}(M_n > (1 + \epsilon)\log n) = 1 - \mathbb{P}(X_1 \leq (1 + \epsilon)\log n)^n = 1 - (1 - n^{-(1+\epsilon)})^n \rightarrow 0, \quad (2.55)$$

as $n \rightarrow \infty$.

2.6 L^p Spaces

October 6th.

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For $0 < p < \infty$, define the set

$$V_p = \{X : \Omega \rightarrow \mathbb{R} \mid \mathbb{E}[|X|^p] < \infty\} \quad (2.56)$$

and (for $p = \infty$) define

$$V_\infty = \{X : \Omega \rightarrow \mathbb{R} \mid \exists M > 0 \text{ such that } \mathbb{P}(|X| > M) = 0\}. \quad (2.57)$$

Proposition 2.14. V_p and V_∞ are real vector spaces with respect to the obvious operations of addition and scalar multiplication.

Proof. For $p \geq 1$, let $X, Y \in V_p$ and $\alpha \in \mathbb{R}$. Then, by the triangle inequality and convexity of $x \mapsto x^p$, we have

$$\mathbb{E}[|X + Y|^p] \leq 2^{p-1}(\mathbb{E}[|X|^p] + \mathbb{E}[|Y|^p]) < \infty, \quad (2.58)$$

and

$$\mathbb{E}[|\alpha X|^p] = |\alpha|^p \mathbb{E}[|X|^p] < \infty. \quad (2.59)$$

Hence V_p is closed under addition and scalar multiplication. For $p < 1$, we use the fact that $|x + y|^p \leq |x|^p + |y|^p$, and the fact that $|\alpha x|^p = |\alpha|^p |x|^p$. The proof for V_∞ is similar. $\blacksquare$

We now wish to define a norm on V_p and V_∞ ; the obvious candidate is to define $\|X\|_p = \mathbb{E}[|X|^p]^{1/p}$ for $0 < p < \infty$, and $\|X\|_\infty = \inf\{M > 0 \mid \mathbb{P}(|X| > M) = 0\}$. However, this is not a norm since $\|X\|_p = 0$ implies $X = 0$ only almost surely and not exactly. The definiteness property fails. Hence, to rectify this, we define an equivalence relation on V_p and V_∞ as follows. For $X, Y \in V_p$, we say that $X \sim Y$ if $\mathbb{P}(X \neq Y) = 0$, and similarly for $X, Y \in V_\infty$. Then, we define the quotient spaces

$$L^p = V_p / \sim \quad \text{and} \quad L^\infty = V_\infty / \sim. \quad (2.60)$$

The norms $\|\cdot\|_p$ and $\|\cdot\|_\infty$ are now definite on L^p and L^∞ , respectively. For $p < 1$, the triangle inequality still fails. Hence, L_∞ and L_p for $p \geq 1$ are normed vector spaces. In fact, these are Banach spaces via the Riesz-Fischer theorem, which states that L^p and L^∞ are complete with respect to their respective norms. We will assume this theorem without proof. The L^2 space is particularly important, since it is a Hilbert space with respect to the inner product $\langle X, Y \rangle = \mathbb{E}[XY]$.

Proposition 2.15. *Let $1 \leq p \leq q \leq \infty$. Then $L^q \subseteq L^p$.*

Proof. If $q = \infty$, then $|X| \leq \|X\|_\infty$ almost surely, and hence $X \in L^p$. Now suppose that $q < \infty$ and let $X \in L^q$. If $p = q$, the result is immediate. For $p < q$, we have

$$\mathbb{E}[|X|^p] = \mathbb{E}[|X|^p \mathbf{1}_{|X| \leq 1}] + \mathbb{E}[|X|^p \mathbf{1}_{|X| > 1}] \leq 1 + \mathbb{E}[|X|^q \mathbf{1}_{|X| > 1}] \leq 1 + \mathbb{E}[|X|^q] < \infty, \quad (2.61)$$

and hence $X \in L^p$. ■

To show triangle inequality for L^p , $p \geq 1$, we require convex functions.

Definition 2.16. A function $\varphi : (a, b) \rightarrow \mathbb{R}$ is a *convex function* if φ is continuous and given any $u \in (a, b)$, there exists a linear function $L_u(x) = \alpha_u x + \beta_u$ such that $L_u(u) = \varphi(u)$ and $L_u(x) \leq \varphi(x)$ for all $x \in (a, b)$. In other words, there exists a line passing through $(u, \varphi(u))$ that lies below the graph of φ .

Lemma 2.17 (*Jensen's inequality*). *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function, and suppose $X, \varphi(X) \in L^1$. Then $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$.*

Proof. Set $a = \mathbb{E}[X]$. Then, by the definition of convexity, there exists a linear function $y = m(x - a) + \varphi(a)$ such that $y \leq \varphi(x)$ for all $x \in \mathbb{R}$, or $m(X - a) + \varphi(a) \leq \varphi(X)$. Taking expectations gives

$$\varphi(\mathbb{E}[X]) = \varphi(a) \leq \mathbb{E}[\varphi(X)]. \quad (2.62)$$

Lemma 2.18 (*Minkowski's inequality*). *Let $1 \leq p < \infty$. Then for any $X, Y \in L^p$, we have*

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p. \quad (2.63)$$

Proof. Set $X' = X / \|X\|_p$ and $Y' = Y / \|Y\|_p$. Then $\|X'\|_p = \|Y'\|_p = 1$. Also set $a = \frac{\|X\|_p}{\|X\|_p + \|Y\|_p}$ and $b = \frac{\|Y\|_p}{\|X\|_p + \|Y\|_p}$. Since $x \mapsto x^p$ is convex, we have

$$|aX' + bY'|^p \leq a|X'|^p + b|Y'|^p \implies \mathbb{E}[|aX' + bY'|^p] \leq a\mathbb{E}[|X'|^p] + b\mathbb{E}[|Y'|^p] = a + b = 1. \quad (2.64)$$

Rearranging gives

$$\begin{aligned} \|X + Y\|_p &= (\mathbb{E}[|X + Y|^p])^{1/p} = (\mathbb{E}[(a + b)(X' + Y')^p])^{1/p} \\ &= (a + b)(\mathbb{E}[|aX' + bY'|^p])^{1/p} \leq a + b = \|X\|_p + \|Y\|_p. \end{aligned} \quad (2.65)$$

Theorem 2.19 (*Hölder's inequality*). *Let $1 < p, q < \infty$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then for any $X \in L^p$ and $Y \in L^q$, we have $XY \in L^1$ and*

$$\mathbb{E}[|XY|] \leq \|X\|_p \|Y\|_q. \quad (2.66)$$

Note that the case $p = q = 2$ is a special case of this theorem, and is known as the Cauchy-Schwarz inequality.

Proof. We first show that $\frac{a^p}{p} + \frac{b^q}{q} \geq ab$ for all $a, b > 0$. To see this, note that the function $f(a) = \frac{a^p}{p} - ab + \frac{b^q}{q}$ is convex in a , and has a unique minimum at $a = b^{1/(p-1)}$. Evaluating f at this point gives

$$f(b^{1/(p-1)}) = \frac{b^{p/(p-1)}}{p} - b^{p/(p-1)} + \frac{b^q}{q} = b^q \left(\frac{1}{p} - 1 + \frac{1}{q} \right) = 0. \quad (2.67)$$

This shows that $\frac{a^p}{p} + \frac{b^q}{q} \geq ab$ for all $a, b > 0$. Now, let $X \in L^p$ and $Y \in L^q$, and set $X' = X/\|X\|_p$ and $Y' = Y/\|Y\|_q$. Then $\|X'\|_p = \|Y'\|_q = 1$. Applying the inequality above with $a = |X'|$ and $b = |Y'|$, we have

$$|X'Y'| \leq \frac{|X'|^p}{p} + \frac{|Y'|^q}{q} \implies \mathbb{E}[|X'Y'|] \leq \frac{\mathbb{E}[|X'|^p]}{p} + \frac{\mathbb{E}[|Y'|^q]}{q} = \frac{1}{p} + \frac{1}{q} = 1. \quad (2.68)$$

Multiplying both sides by $\|X\|_p \|Y\|_q$ gives the desired result. ■

With the norm defined on L^p , we can define convergence in L^p by saying that a sequence $(X_n)_{n \geq 1}$ converges to X in L^p if $\|X_n - X\|_p \rightarrow 0$ as $n \rightarrow \infty$.

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